

20
27

BEDIRHAN CAKIROGLU

Portfolio

INDUSTRIAL DESIGNER

IND. DESIGN ENGINEER

bedirhancakiroglu.com

Bedirhan Çakıroğlu

Industrial Designer · Design Engineer

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bedirhancakiroglu.com
linkedin.com/in/bedirhanckr
Paderborn, Germany



TECHNICAL SKILLS

CAD HiCAD, SolidWorks, Rhino, Fusion 360

Rendering KeyShot, V-Ray, Blender

Adobe Photoshop, Illustrator

Prototyping 3D Printing, Mock-Up Making, Hand Sketching

EDUCATION

B.Sc. Industrial Design 2018 – 2023
Istanbul Technical University GPA 3.46 / 4.00

Specialized in user-centered product development, product sketching, physical prototyping and CAD modeling. Developed multiple design projects from initial research and ideation to manufacturing preparation.

LANGUAGES

Turkish Native
English C1
German B1–B2 (in progress)

LICENSE & INTERESTS

License A2 & B class

Interests Motorcycling · Swimming · Paddleboarding · Cooking · Darts · Traveling

PROFESSIONAL SUMMARY

Industrial Designer and Design Engineer with experience developing products from concept to production. Skilled in CAD modeling, technical drawings, prototyping, and design for manufacturing. Comfortable working across design, engineering, and production.

WORK EXPERIENCE

Industrial Designer & Design Engineer Jan 2024 – Present
TUS Metallbau Paderborn, Germany

- Over 150 steel-construction designs realized in production since 2023.
- Design and develop balcony systems, canopies, railings, steel structures and glass facades, ensuring compliance with local engineering and building standards.
- Create 3D models, assemblies and production drawings; formulate BOMs and support material procurement and production scheduling.
- Carry out on-site measurements and incorporate site constraints into CAD models; optimize designs for manufacturing methods, tolerances and assembly.

Industrial Design Intern Jun 2023 – Dec 2023
TUS Metallbau Paderborn, Germany

Design Engineer Aug 2022 – Jun 2023
ITU ROV Team Istanbul, Turkey

- Designed watertight enclosures, structural chassis and mechanical mounting systems for ROVs; integrated components with underwater sensors and propulsion systems.
- Designed the external hull of Pioneer MKII, including the nose module and mid-body compartment. Recognized with the TEKNOFEST Design & Presentation Award.
- Represented ITU at the international MATE ROV Competition (Norway), securing 3rd place.

Product Design Intern Feb 2022 – Aug 2022
Robotistan Istanbul, Turkey

- Developed Arduino-based educational robotic kits from concept to physical prototypes; designed injection-molded enclosures, snap-fits and mechanical linkages.
- Created product renderings and technical illustrations for assembly manuals.

HONORS & AWARDS

2023 TEKNOFEST
Design & Presentation Award
Underwater ROV external hull, Pioneer MKII

2023 FURNITUR 2023
Prize & Mention, Türkiye Tasarım Vakfı
Speculative furniture design, future scenarios

2023 Graduation Awards
2nd Prize, Istanbul Technical University
Product & system design for recreation, Rockhounder

2022 Design for Export
Finalist, OAİB & Ministry of Commerce
Multi-functional furniture, Delta

REFERENCES

“Bedirhan stands out for his strong commitment to the design process and his exceptional attention to detail. His sketching skills, prototyping ability, and engineering-oriented approach allow him to translate ideas into well-resolved design outcomes with confidence. This combination of technical rigor, perseverance, and openness to learning makes him a promising industrial designer.”

Dr. Miray Boğa
Assistant Professor, Department of Industrial Design, Iowa State University

Selected Work

01 Pioneer MKII

PRODUCT DESIGN · MECHATRONICS

02 Rockhounder

SOFT GOODS · INDUSTRIAL DESIGN

03 Herptile

PRODUCT & SERVICE DESIGN

04 Peanut Mini Robot

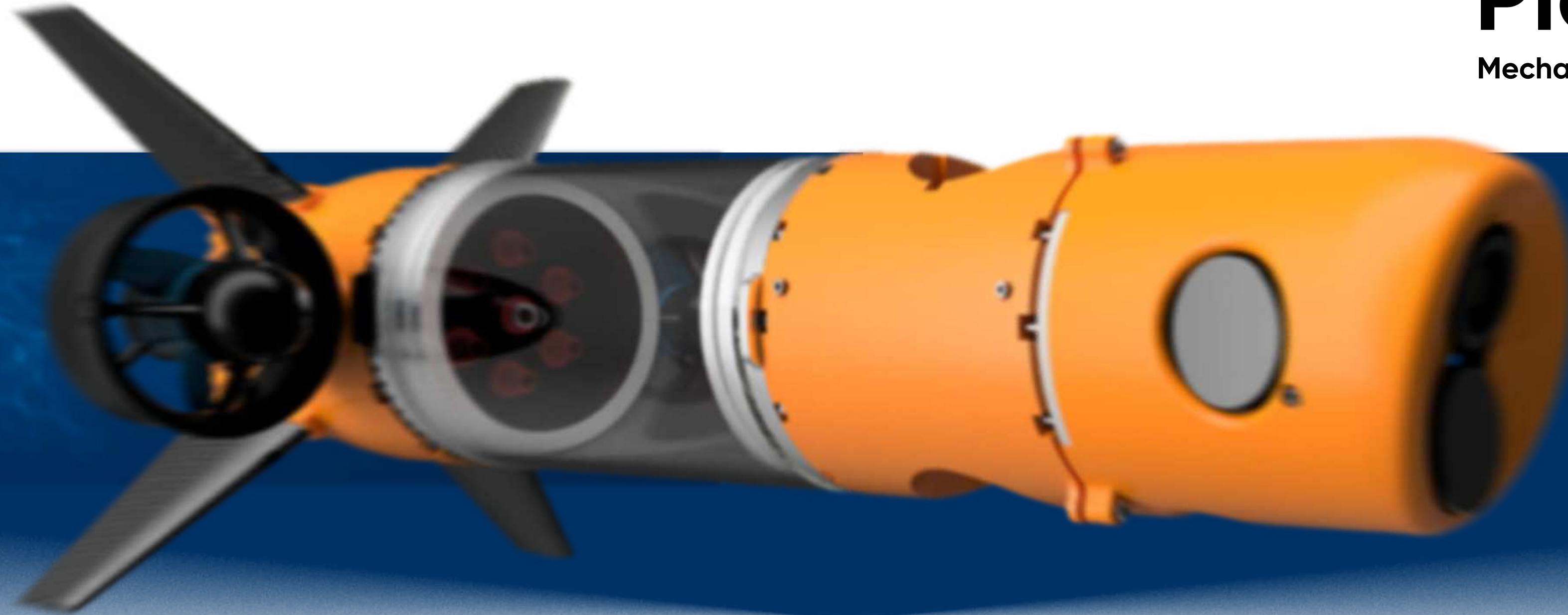
PRODUCT DESIGN · ELECTRONICS

05 Other Selected Work

METAL CONSTRUCTION · ADDITIONAL PROJECTS



01



UNDERWATER VEHICLE REDESIGN

Pioneer MK II

Mechanical Engineering & Product Design

Redesigning Pioneer for improved sensing, serviceability, hydrodynamics and field handling.

ITU ROV Team 2022-2023

Role - Mechanical &
Product Designer

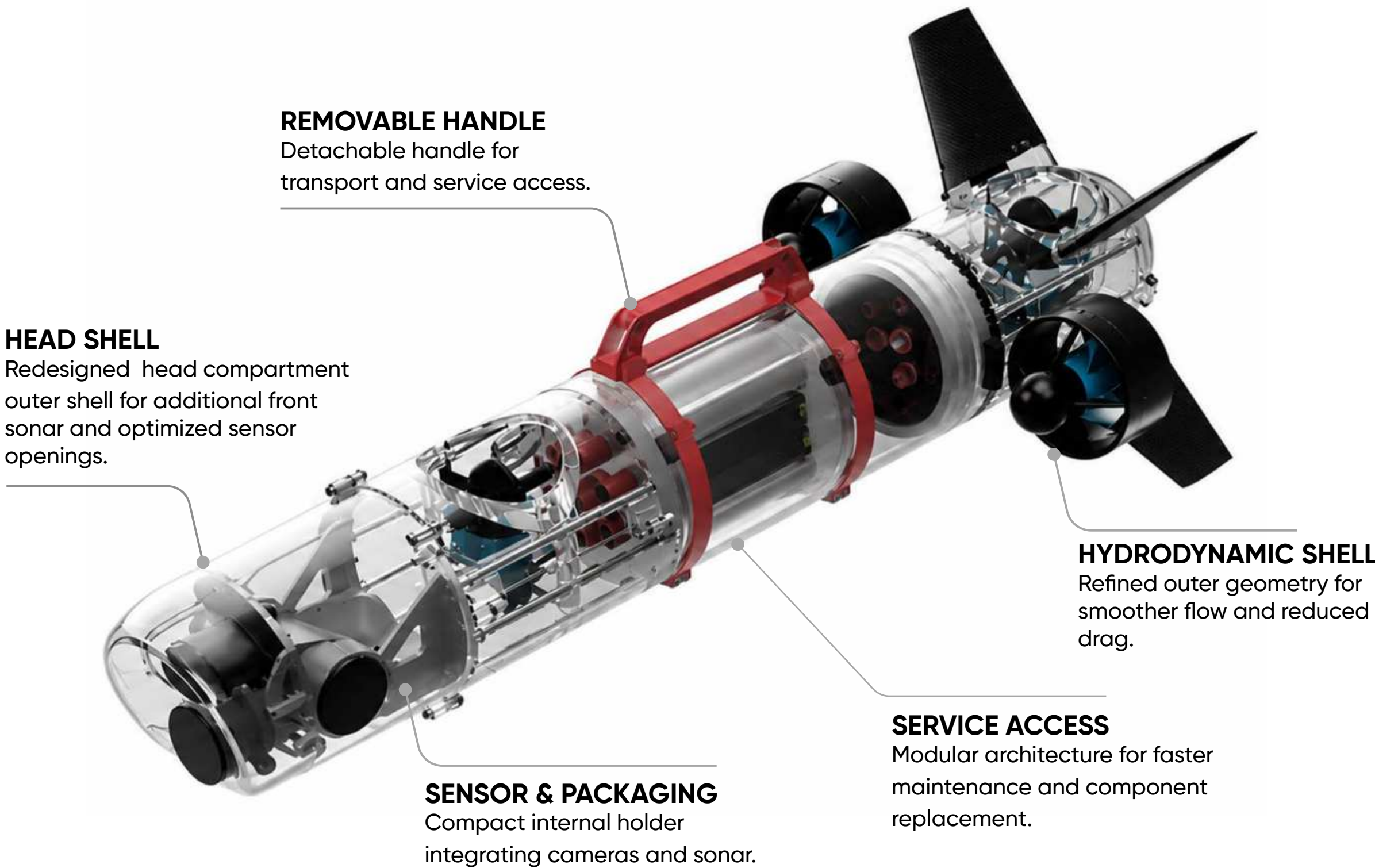
DESIGN BRIEF

From a working platform to a more capable system

The redesign focused on increasing sensing capability while improving the vehicle architecture for maintenance, flow, handling.

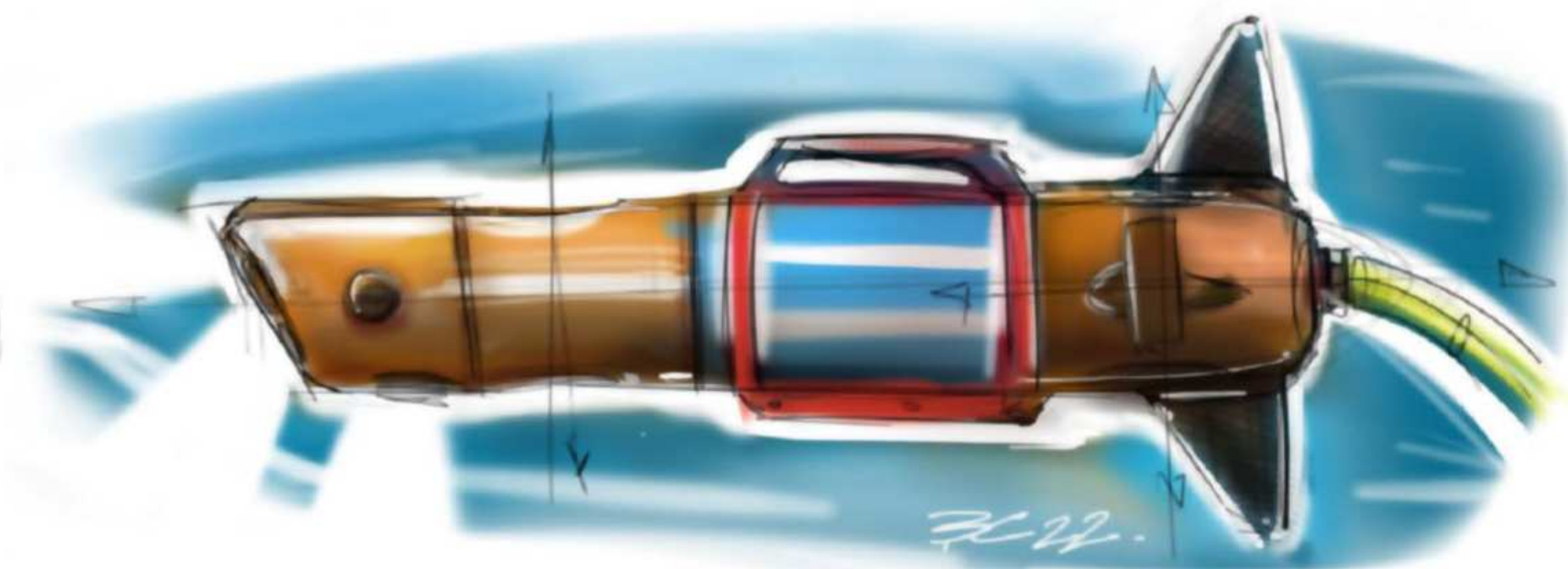
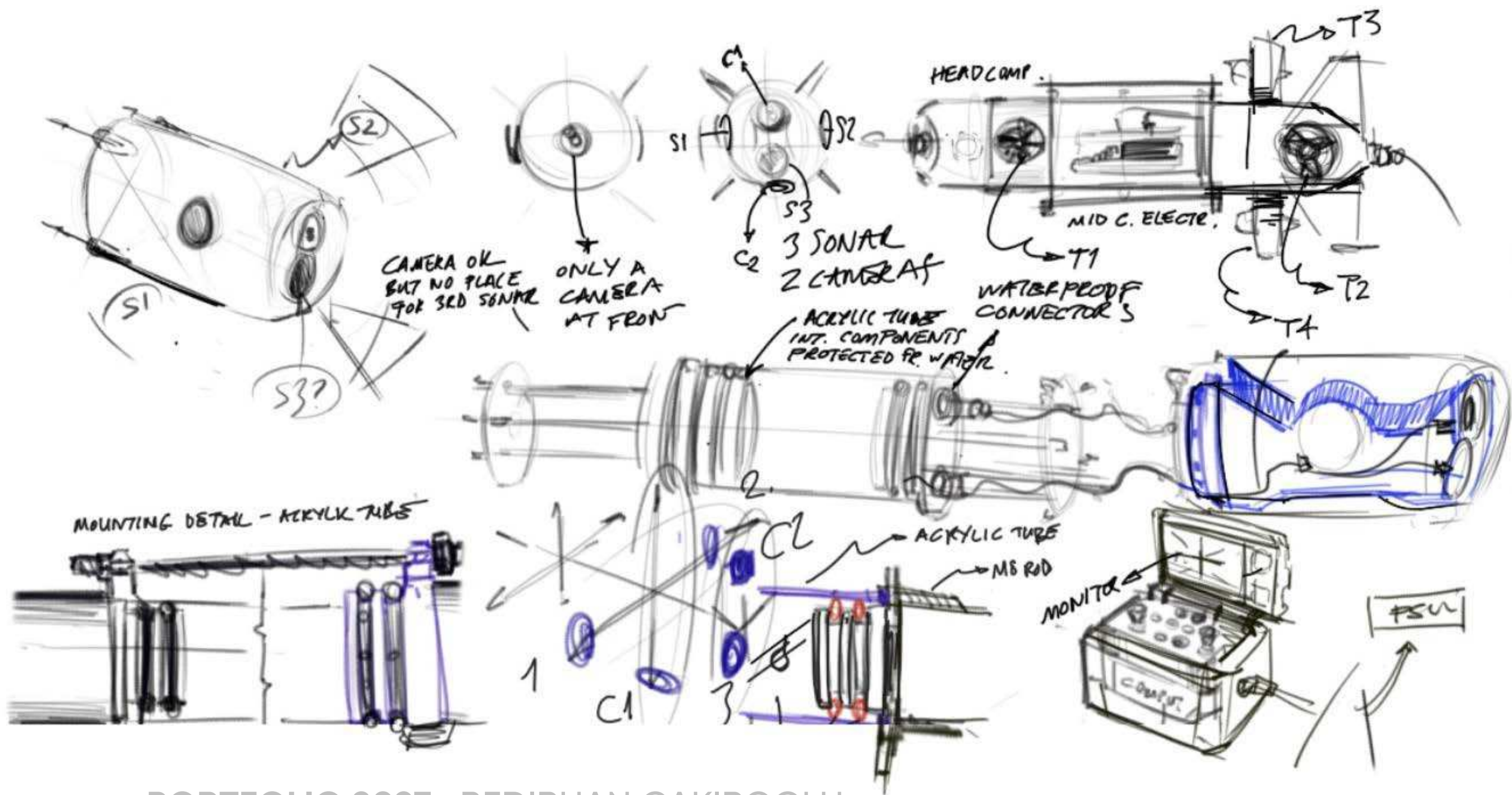
MY CONTRIBUTION

- 01 Sensor Packaging**
Redesigned internal sensor holder for the head compartment to integrate sonar and cameras, including additional front sonar.
- 02 Head Compartment**
Outer shell redesign with optimized sensor openings and hydrodynamic form.
- 03 Structural Design**
Component layout, structural integration and packaging.
- 04 Hydrodynamics**
Refined outer geometry for improved flow and reduced drag.
- 05 Serviceability**
Modular design for easier maintenance, part replacement and transport.



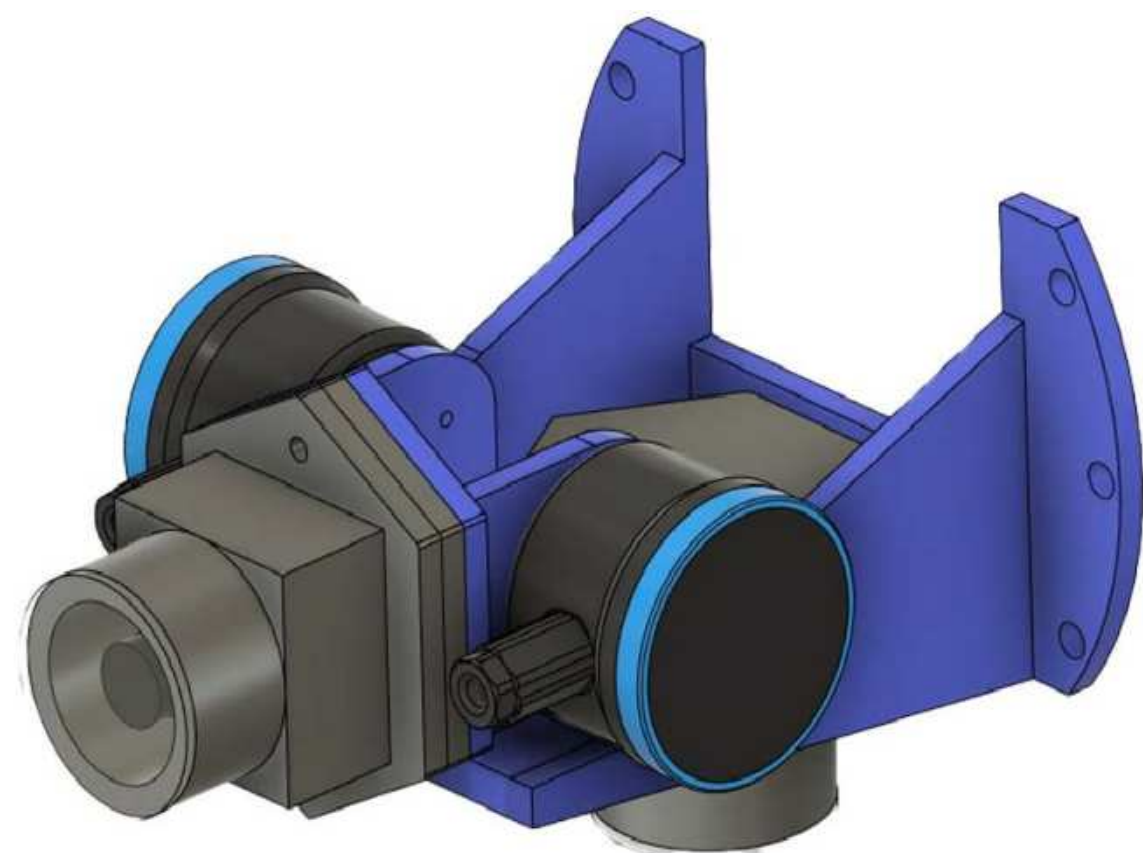
Exploration Sketches

Early studies explored component packaging, shell geometry, sensor integration and service access.



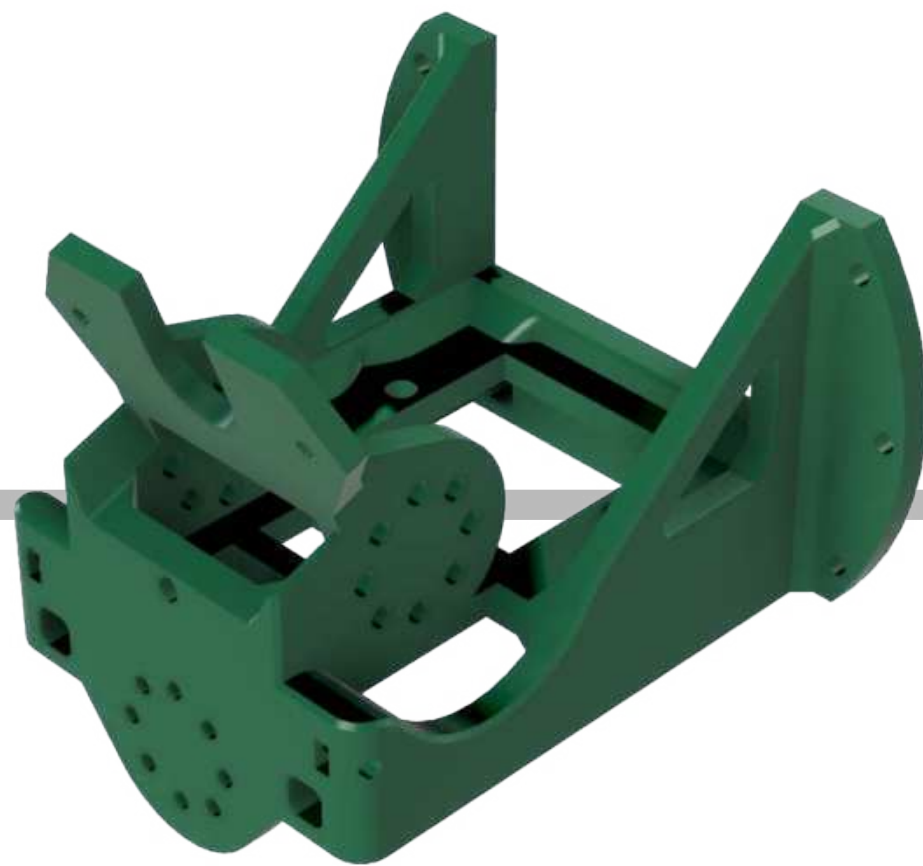
Redesigning the head compartment for expanded sensing

New sensing requirements drove a complete redesign of the internal sensor architecture, integrating two cameras and additional acoustic sensing within the available package.



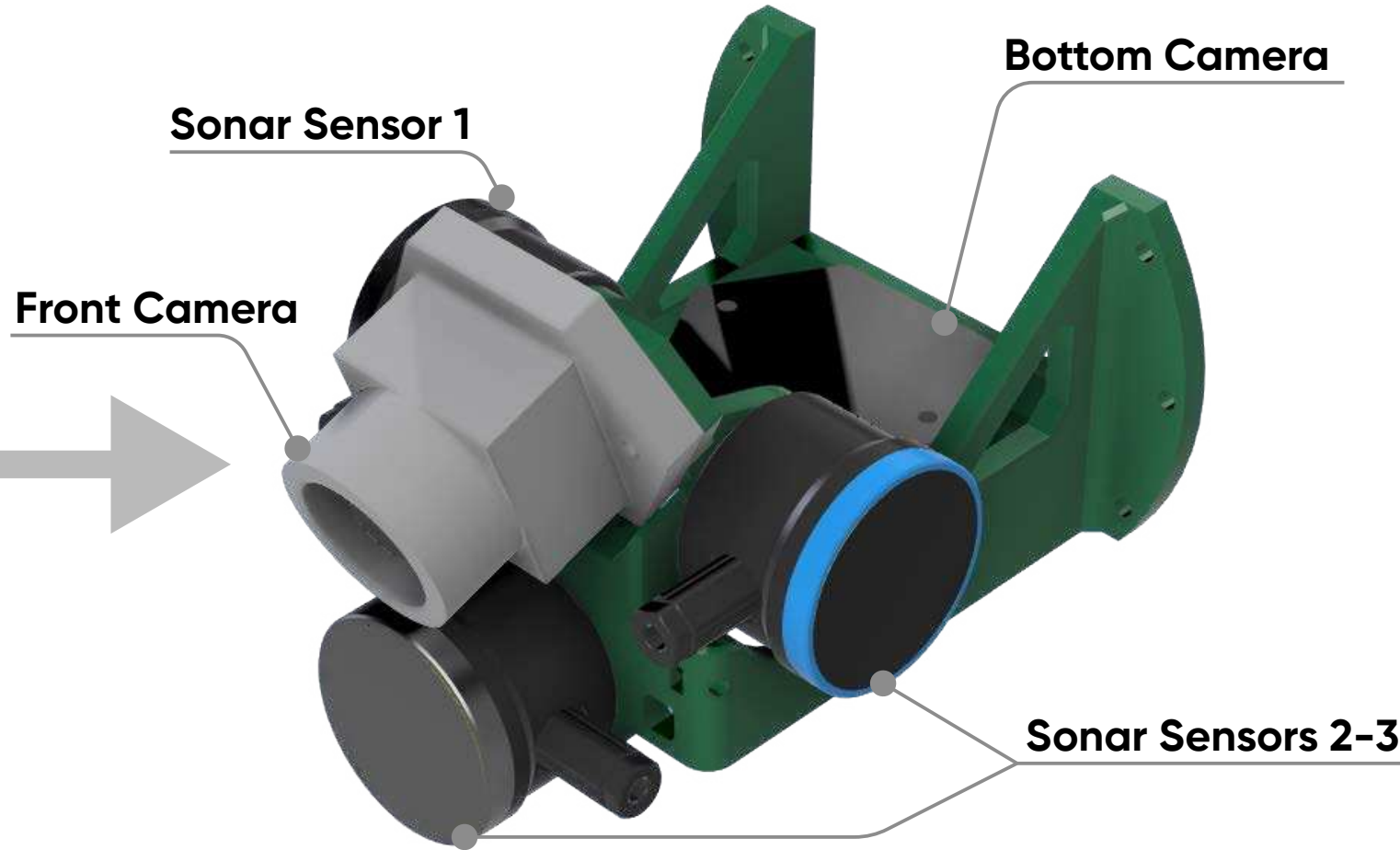
01 _ PREVIOUS ARCHITECTURE

Existing sensor mount with limited packaging space.



02 _ REDESIGNED CARRIER

Compact structural carrier developed



03 _ INTEGRATED SENSOR PACKAGE

Two cameras and acoustic sensors integrated within the head compartment.

INTERNAL ARCHITECTURE

Making the system legible

Transparent studies were used to evaluate packaging relationships between sensing, propulsion and the pressure housing

01 HEAD COMPARTMENT

sensor + camera packaging

02 CENTRAL HOUSING

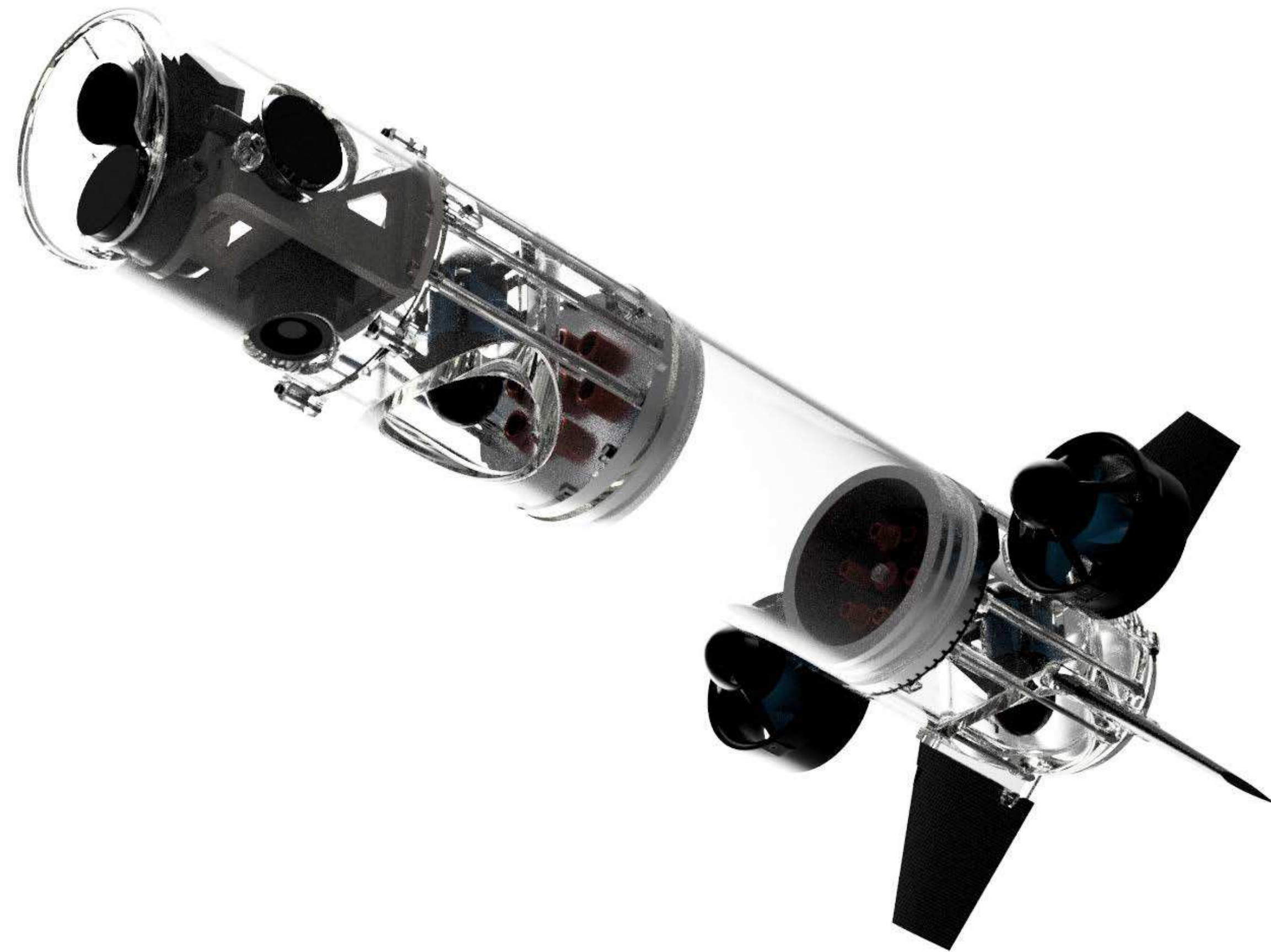
protected system volume

03 THRUSTER ASSEMBLY

maneuvering architecture

04 STRUCTURAL PATH

internalized chassis elements



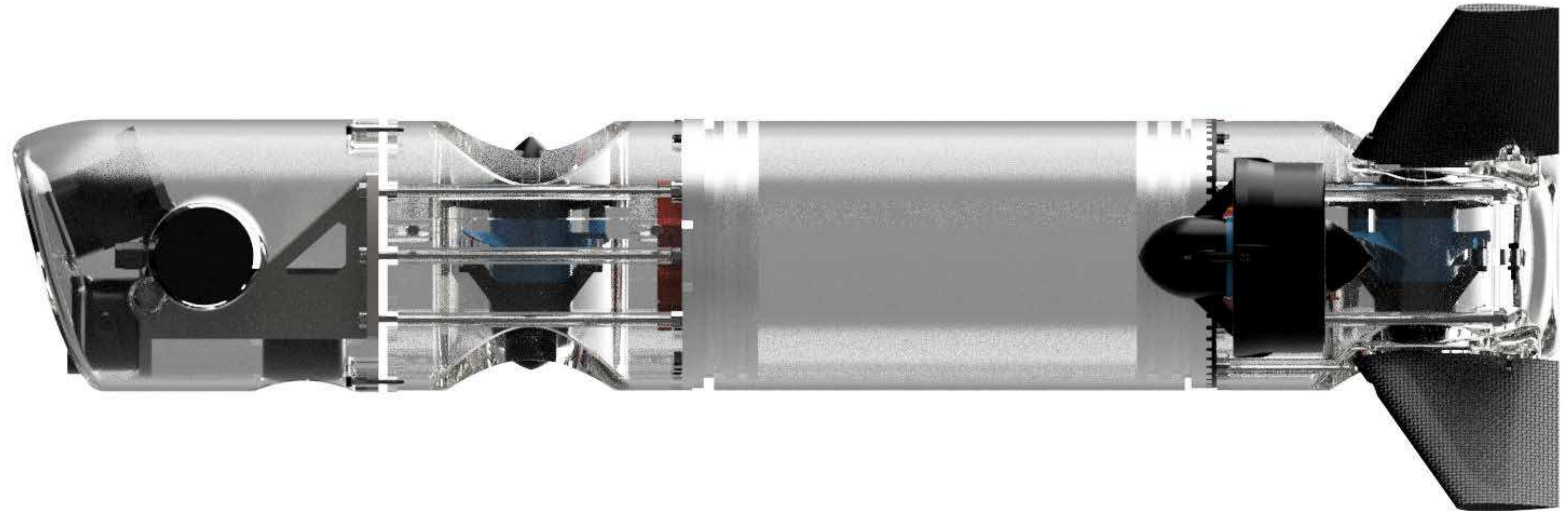
STRUCTURE + FLOW

Cleaning the external envelope

The middle-compartment relationship was reconsidered around rigidity, assembly access and a cleaner flow path.

M8 tie rods moved inside the body envelope

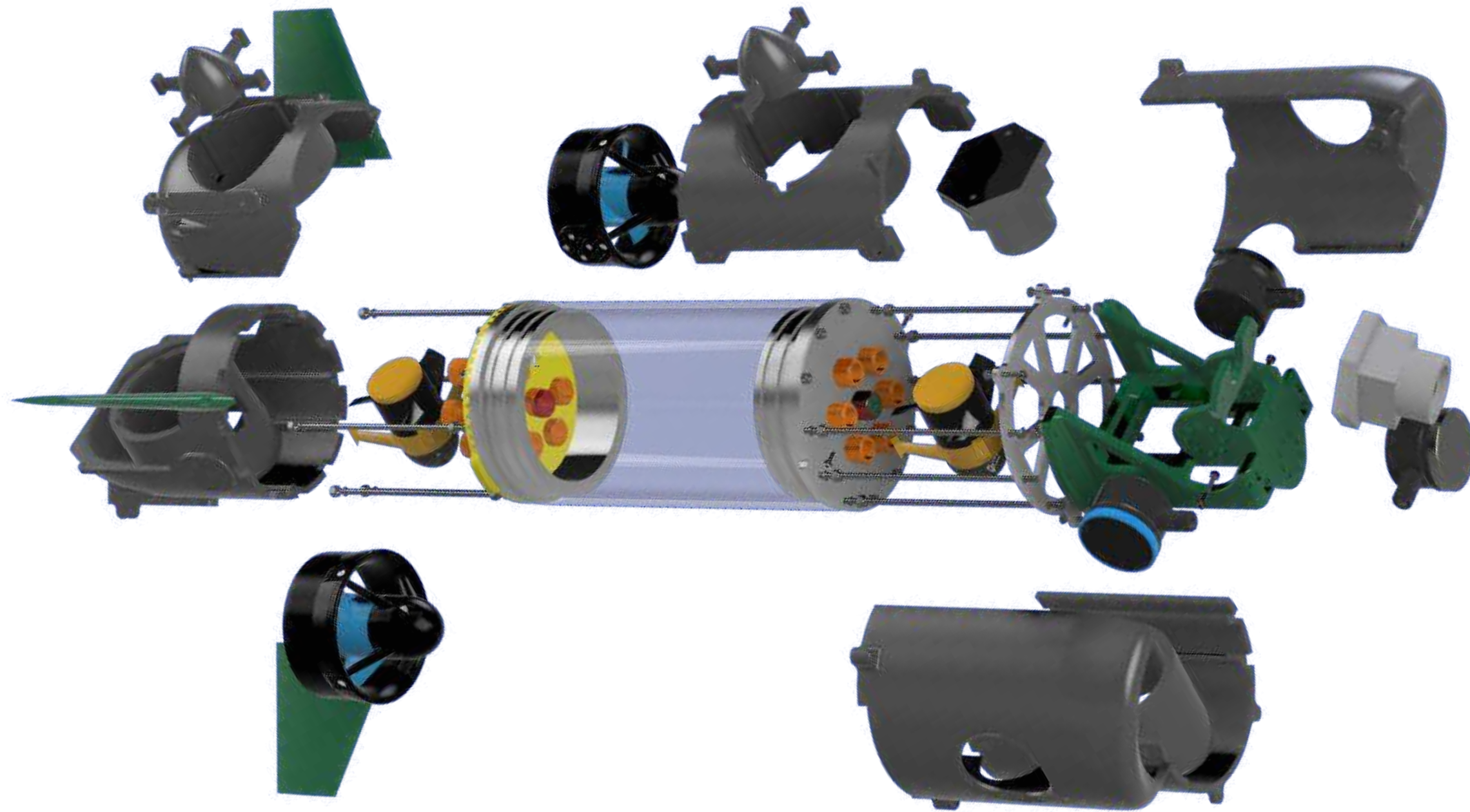
Maintains the structural load path while reducing exposed hardware and creating a cleaner profile.



SERVICEABILITY

Designing for disassembly

The architecture was developed to reduce friction during access, replacement and reassembly of modular sections.



**MODULAR ACCESS
COMPONENT REPLACEMENT
SIMPLER REASSEMBLY**

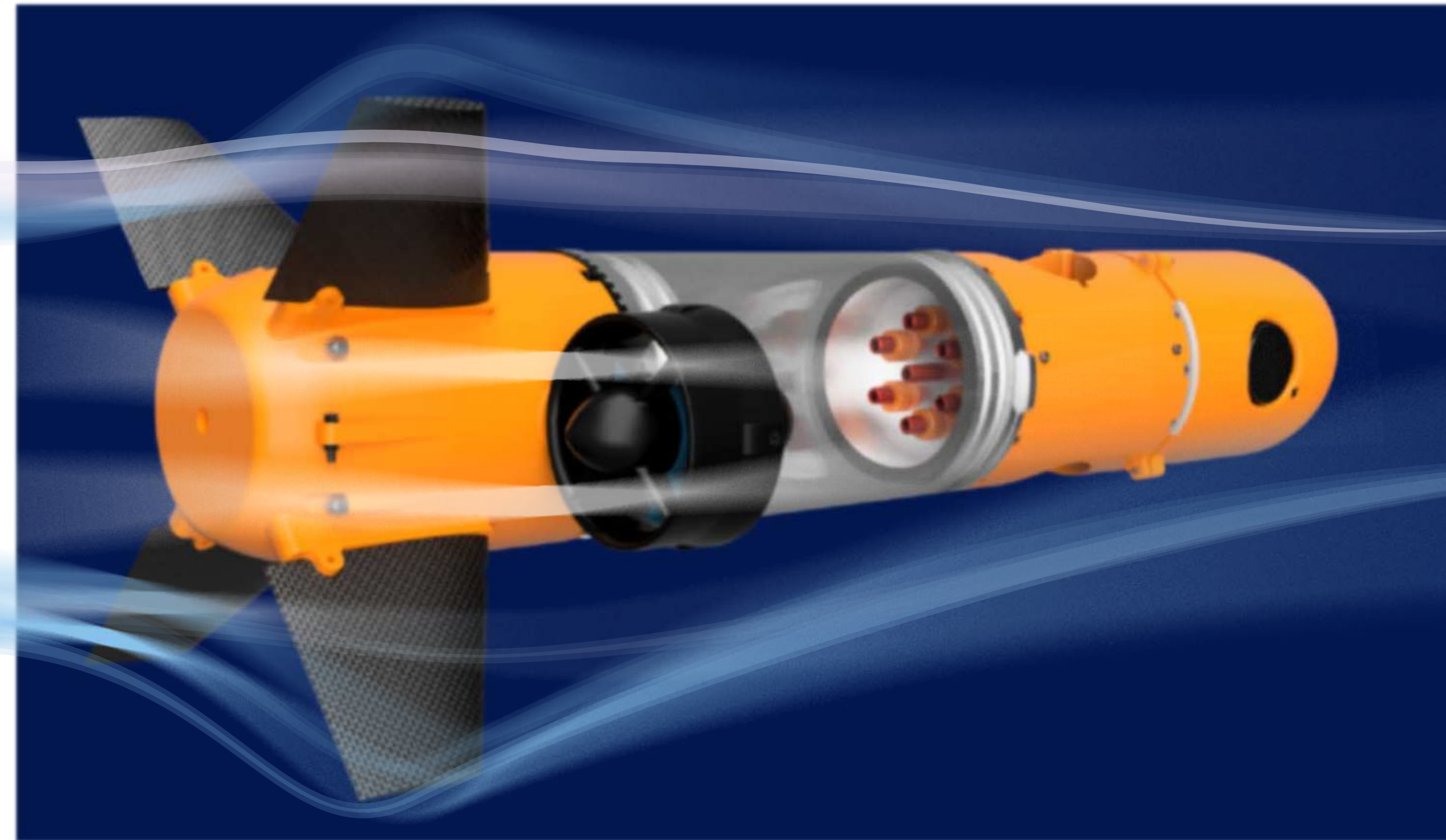
STABILITY & MATERIALS

Lightweight stabilizing surfaces

The tail surfaces were redesigned around stability, stiffness and lower mass.

CARBON FIBER TAILS

Composite fins provide a lightweight and rigid solution for the vehicle's stabilizing surfaces



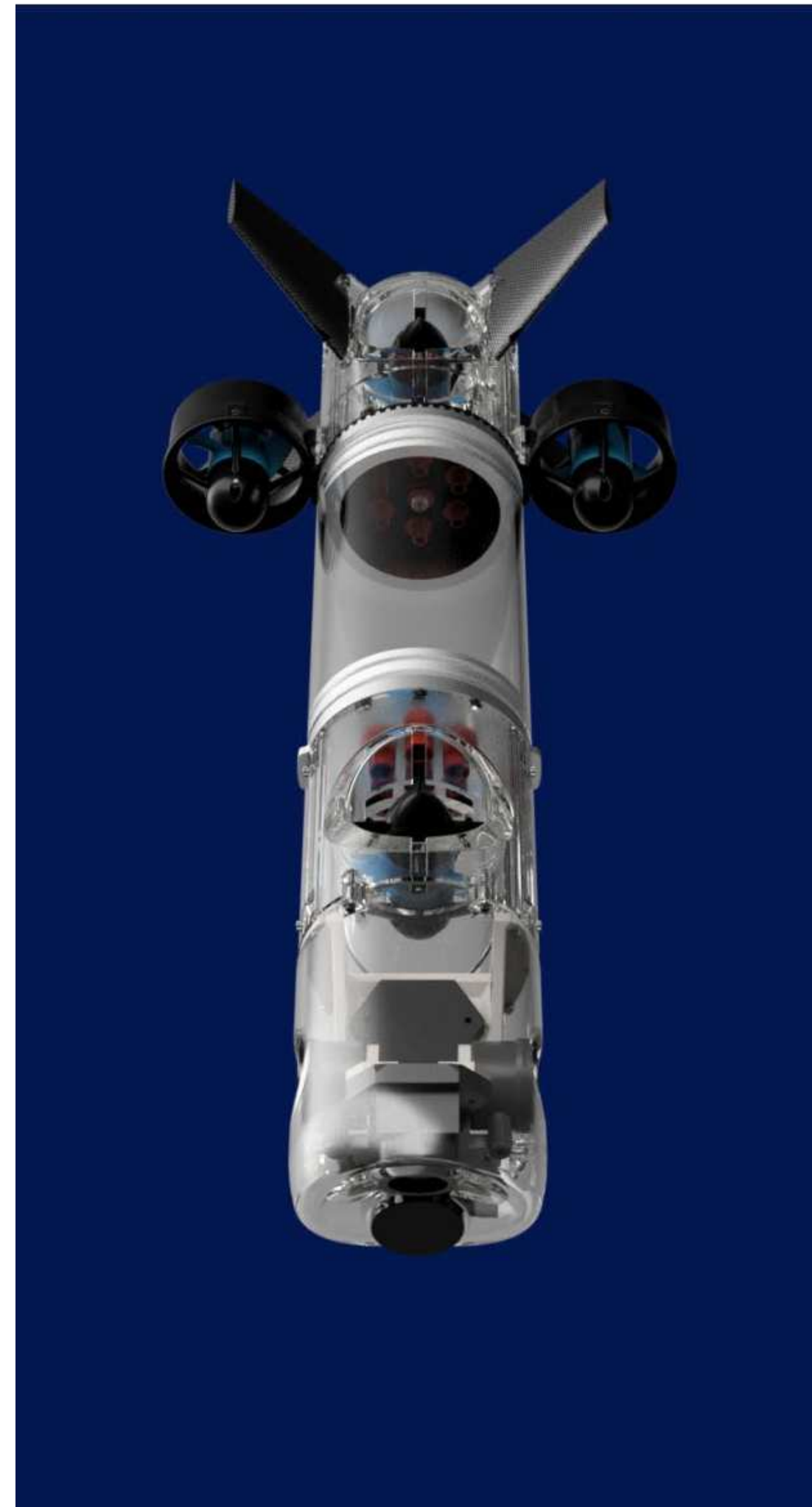
FINAL EXTERIOR

A technical object with a cleaner identity

The final enclosure resolves the internal architecture into a compact, readable external form.

Structure contained within the body

Structural elements and internal components are contained within the enclosure, leaving only the thrusters and stabilizers exposed outside the main body.



ENGINEERING OVERVIEW

Solid outside. Structured inside.

A compact summary of the redesign decisions that define Pioneer MKII.

Sensor integration

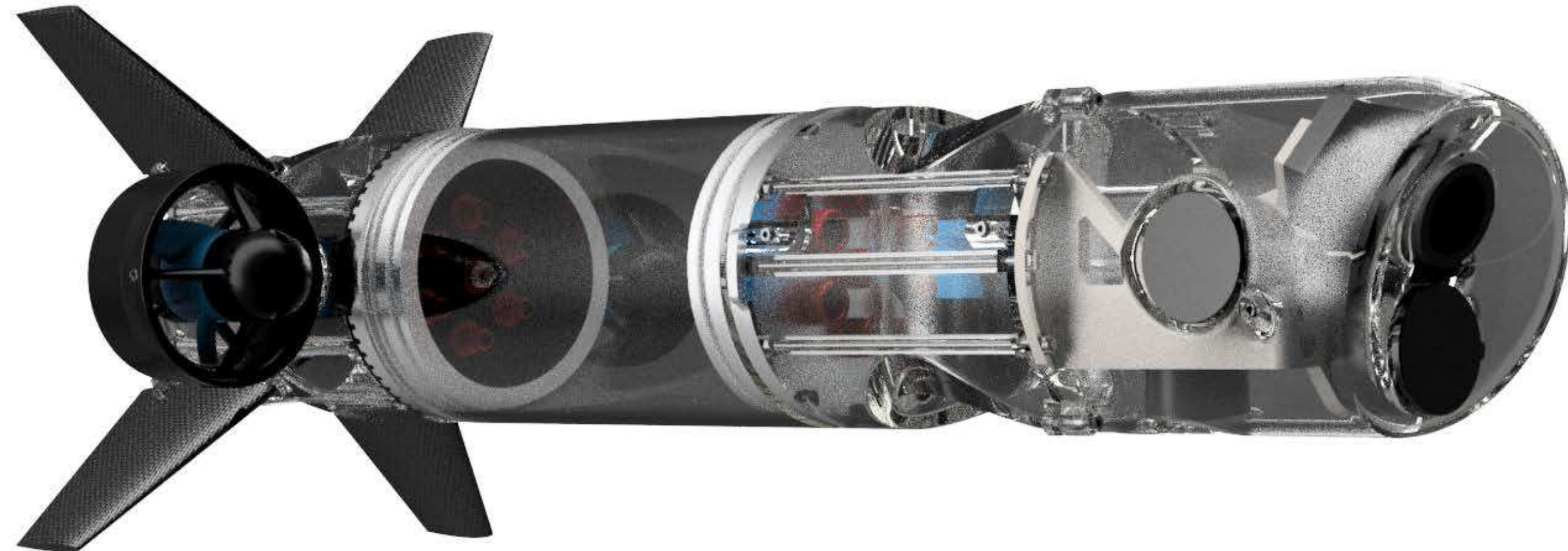
Structural packaging

Serviceability

Hydrodynamic refinement

Composite stabilizers

Handling & transport



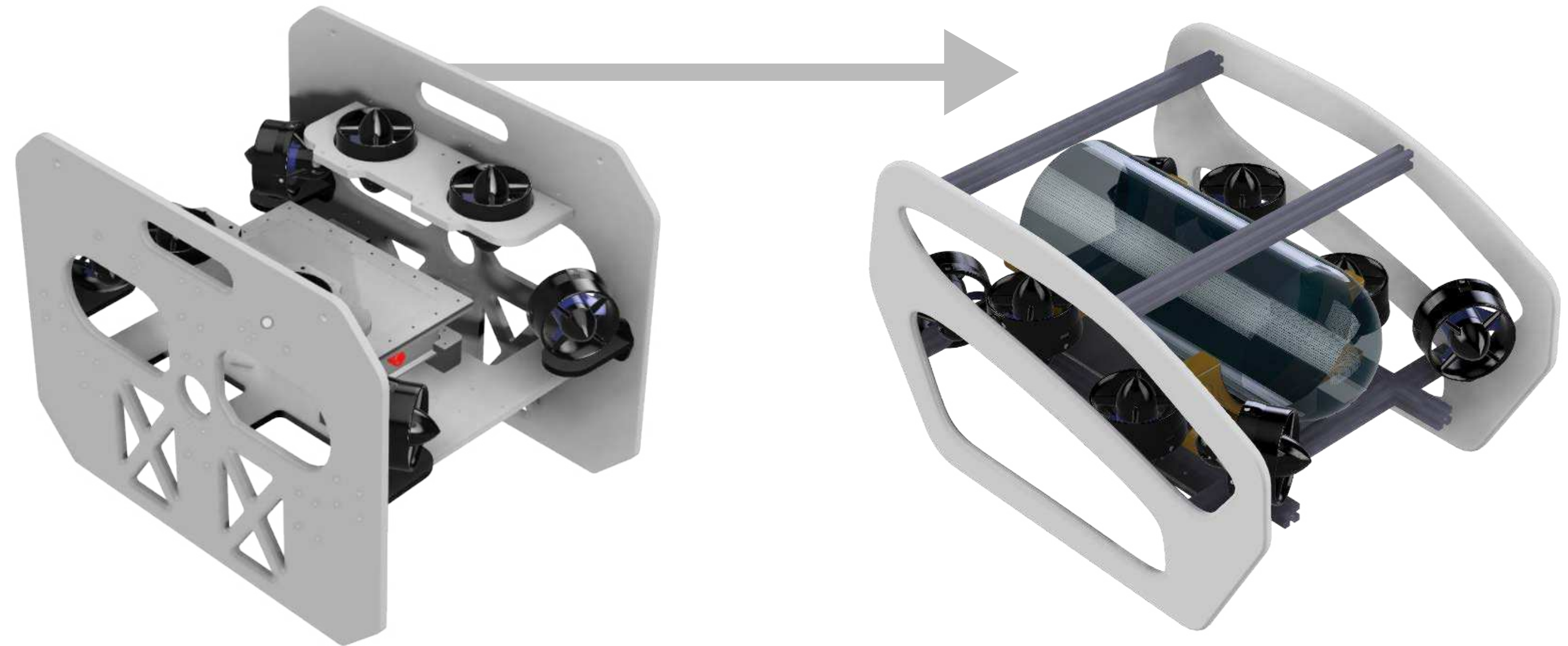
EXPLORER REDESIGN MAIN BODY ASSEMBLY

OTHER WORKS _ ITU ROV TEAM

Additional projects beyond Pioneer MKII.

EXPLORER Underwater Vehicle Project

I contributed to Explorer with design proposals focused on a lightweight, robust and highly maneuverable vehicle architecture, with particular attention to easy assembly and disassembly. I also took an active role in design reviews and team discussions.



ITU ROV TEAM · 2022–2023

One year of designing, building and competing.

A year of hands-on development with the ITU ROV Team – from design and engineering decisions to assembly, testing and competition.



Vehicle: PIONEER MKII
TEKNOFEST – aselsan
Design & Presentation Award

Vehicle: EXPLORER
TAC Challenge · Norway
3rd Place



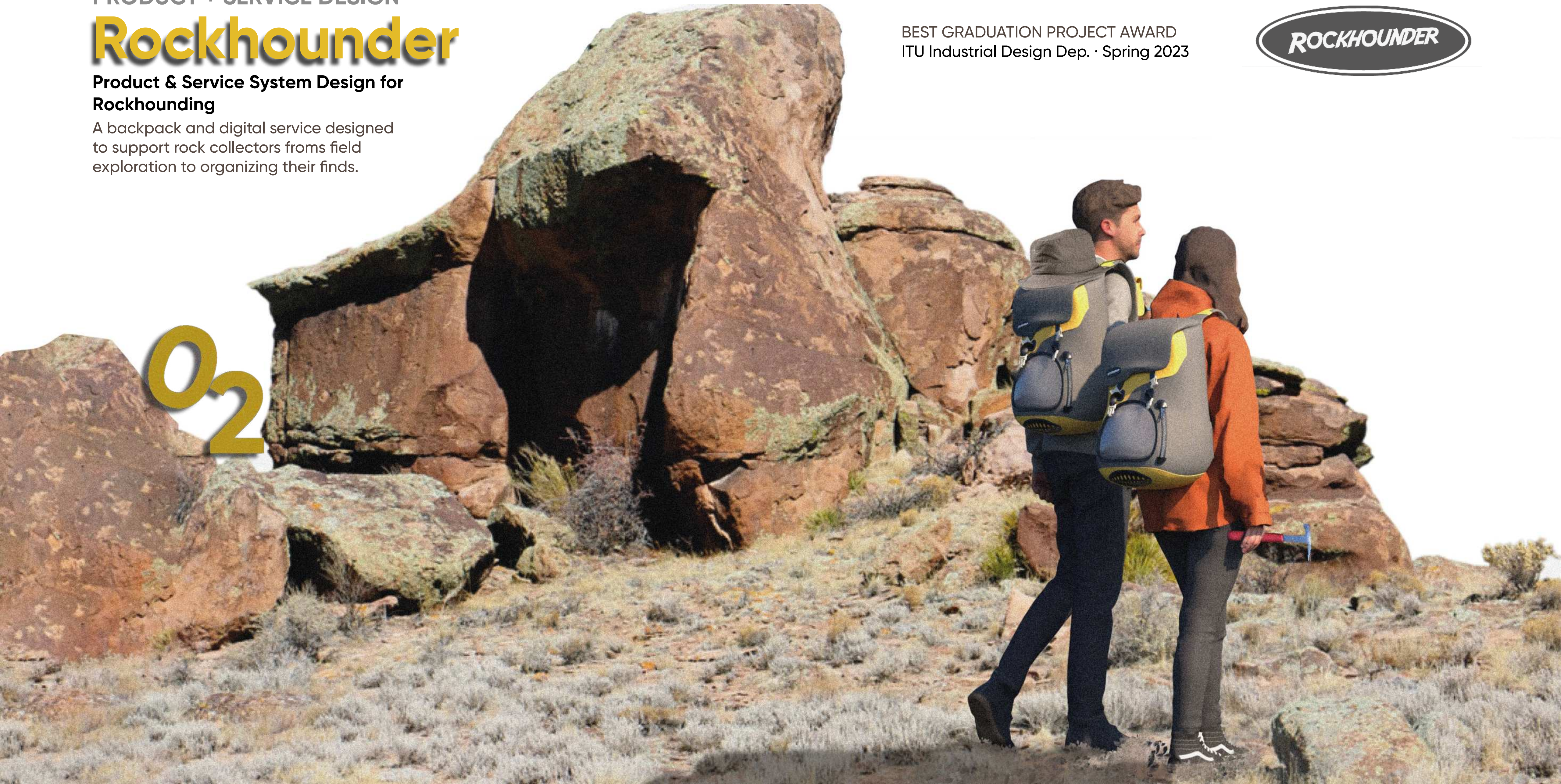
PRODUCT + SERVICE DESIGN

Rockhounder

**Product & Service System Design for
Rockhounding**

A backpack and digital service designed
to support rock collectors froms field
exploration to organizing their finds.

BEST GRADUATION PROJECT AWARD
ITU Industrial Design Dep. · Spring 2023



02

DESIGN BRIEF

Understanding the rockhounding experience

Defining the user, needs and experience



PAIN POINTS

- Difficult to carry dirty and wet specimens.
- Field tools are difficult to organize and access.
- Collected specimens need identification and organization.
- Standard backpacks are not designed around rockhounding activities.

PERSONA

An outdoor enthusiast who enjoys exploring new places and experiences, but prefers guided group activities over planning independent trips.

CONCEPT

A guided weekend experience combining rockhounding and outdoor activities with equipment and planning support.

SYSTEM SCENARIO

1 – DISCOVER & BOOK

Choose and book a guided weekend activity.

2 – ARRIVE & PREPARE

Meet the group and receive equipment and safety guidance.

3 – EXPLORE

Take part in guided outdoor activities.

4 – ROCKHOUND

Collect rocks using dedicated tools and the Rockhounder backpack.

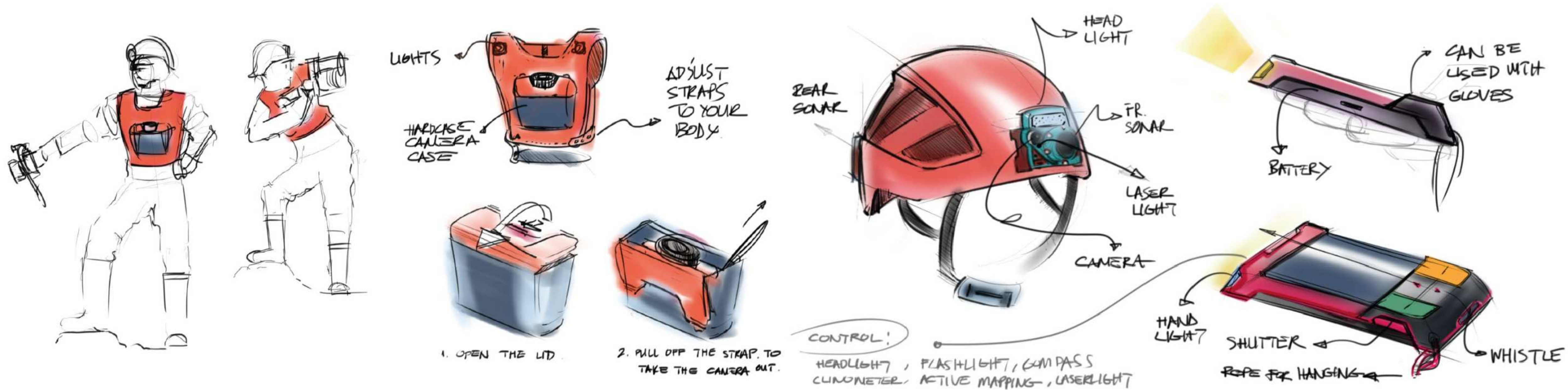
5 – CREATE & CONNECT

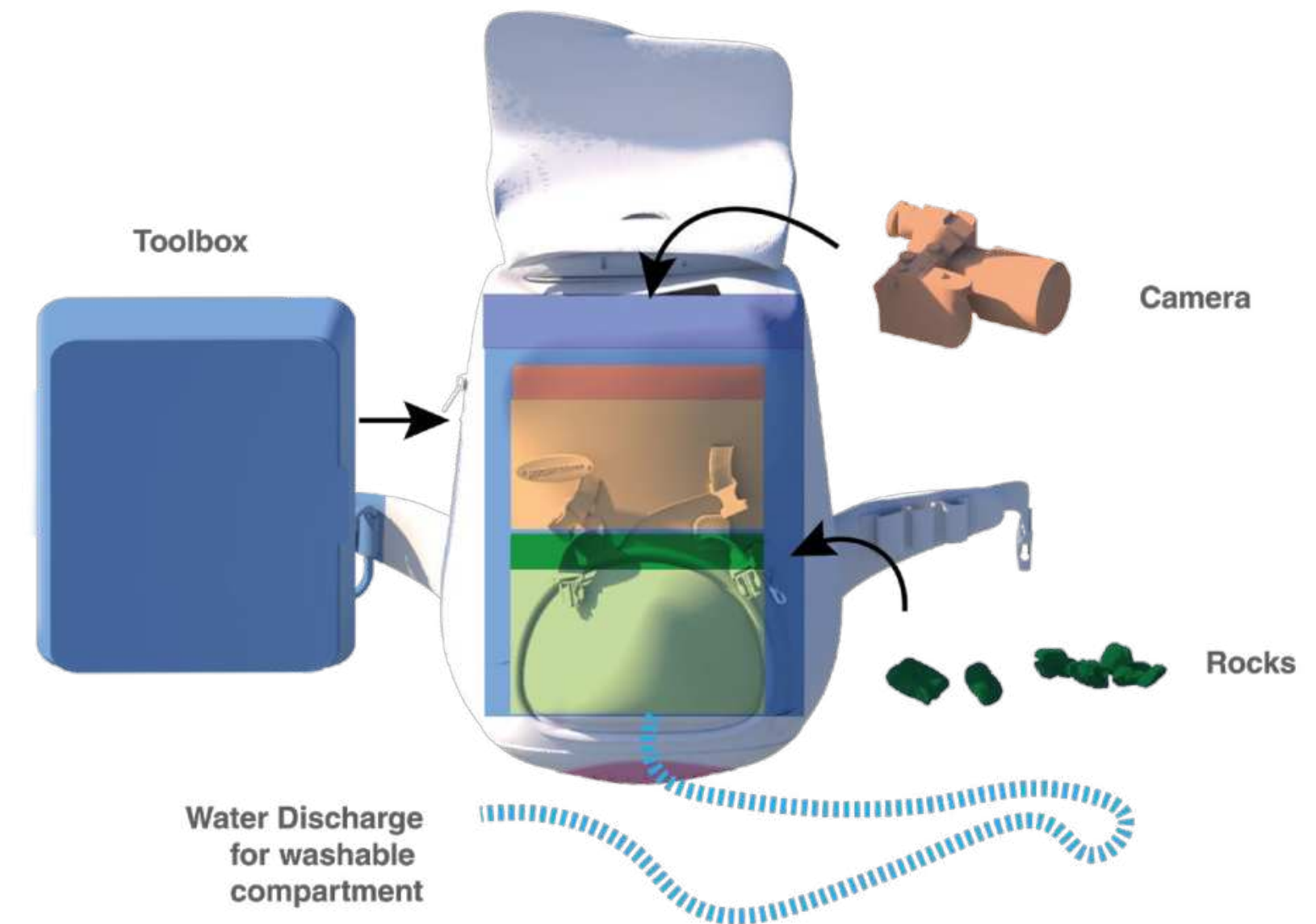
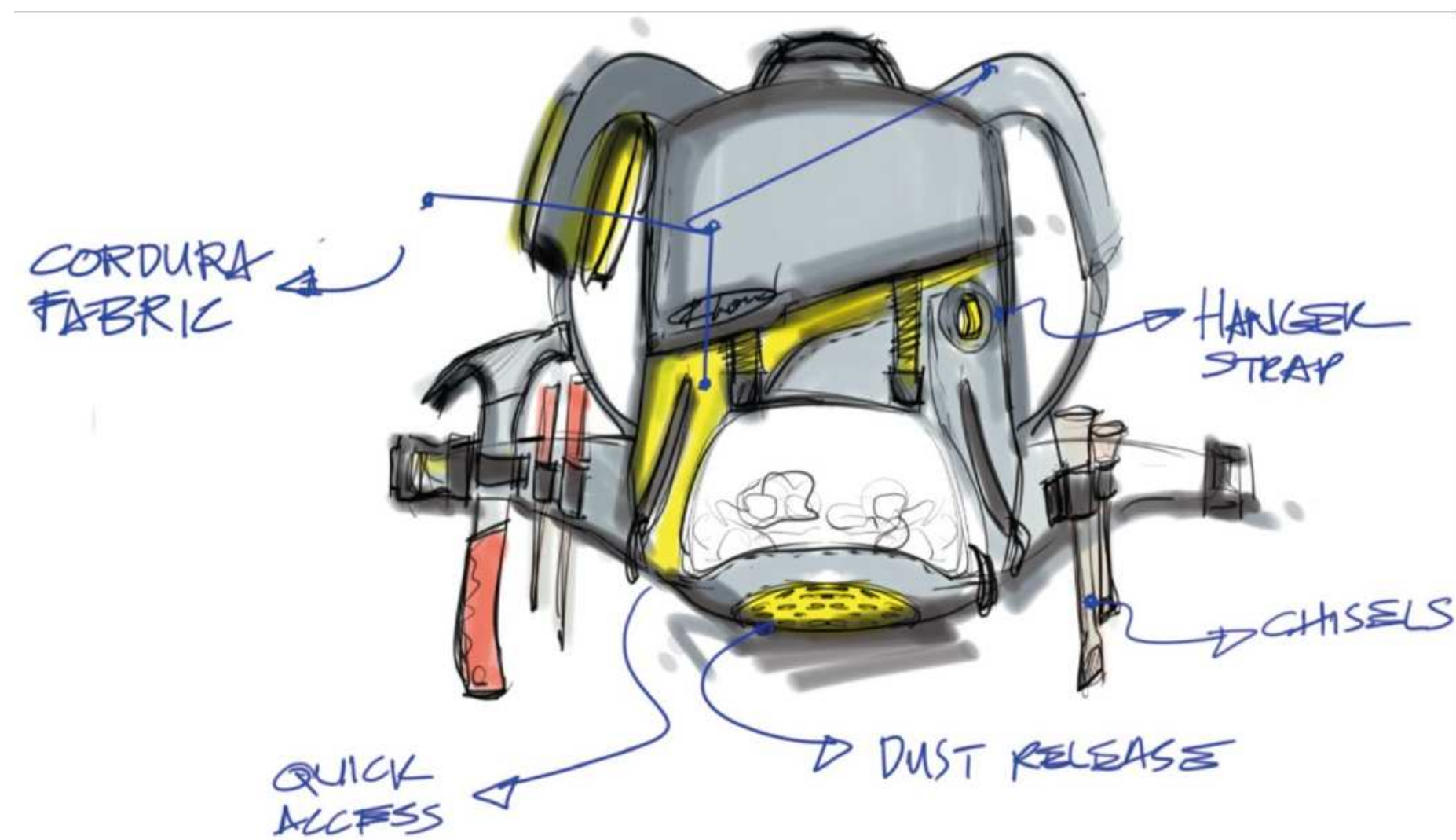
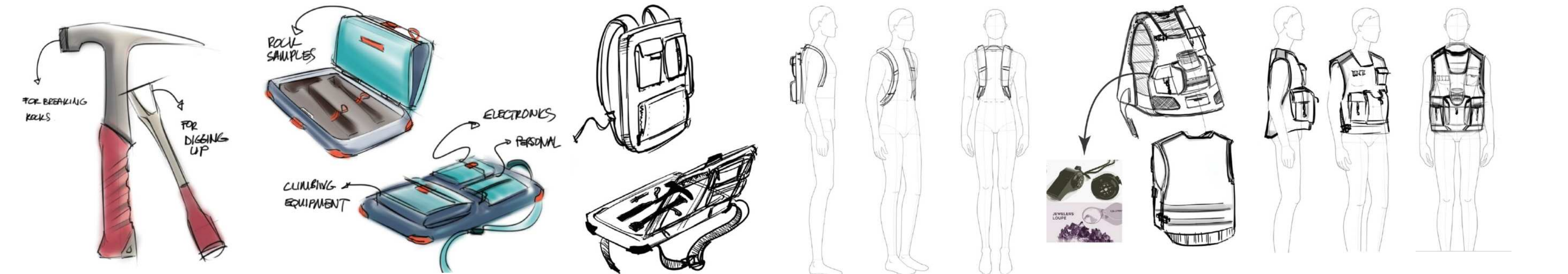
Process the finds at camp and share the experience.

IDEATION & DEVELOPMENT

Exploring the field experience

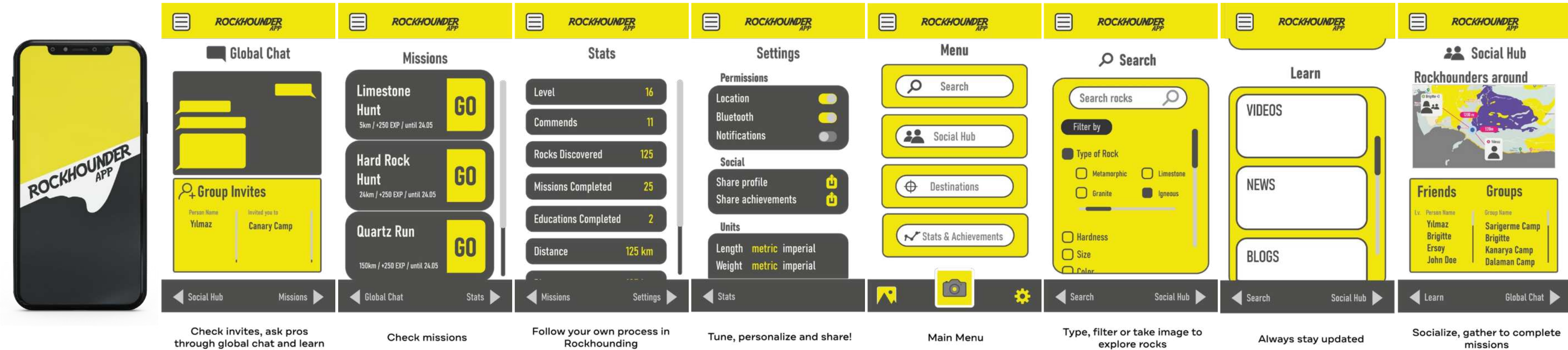
Early concepts explored different ways of carrying equipment, protecting personal gear and supporting outdoor activities.





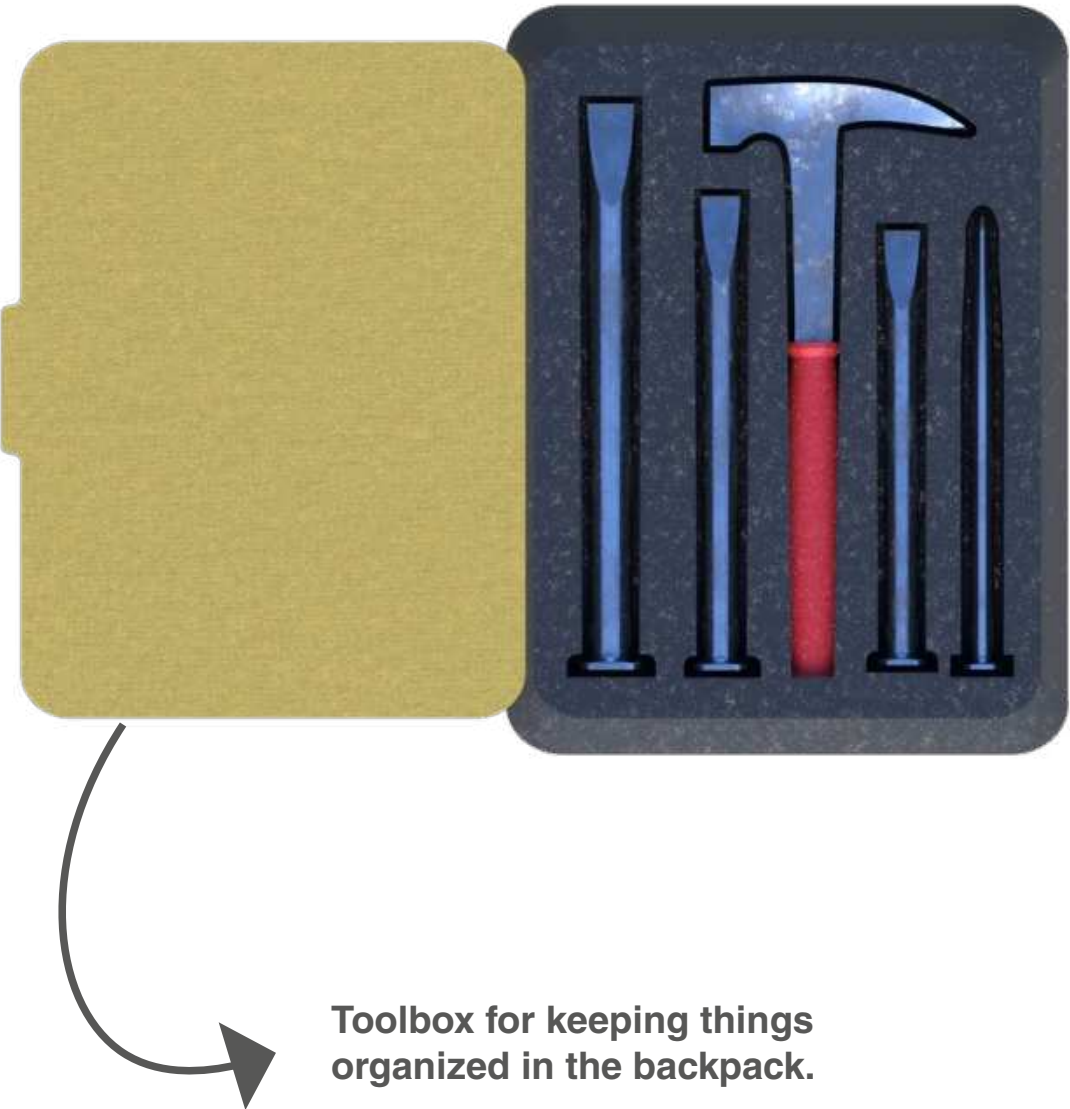
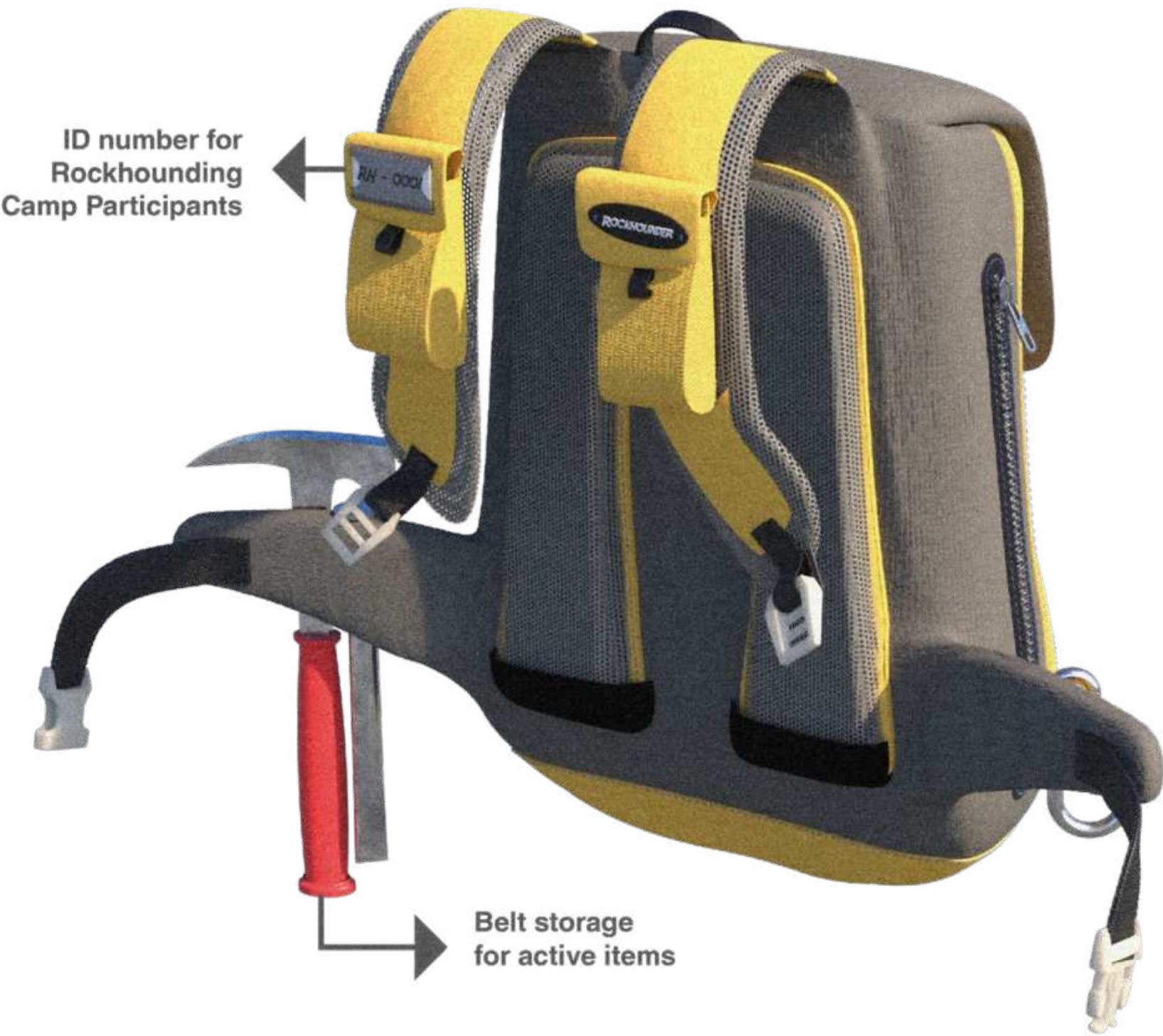
Extending the experience beyond the field

A digital companion designed to help users explore, learn and connect throughout their rockhounding journey.



A Backpack Designed for the Field

Organizing tools, collected rocks and personal equipment around the needs of field exploration.



PHYSICAL PROTOTYPE

From Concept to a Wearable Prototype

Translating the design into a full-scale prototype to evaluate functionality, organization and field use.



1 – PATTERN DEVELOPMENT

Flattening the 3D model into 2D patterns.

2 – PAPER MOCK-UP

Testing proportions and assembly with paper mock-ups.

3– SEWING & ASSEMBLY

Cutting and sewing the fabric based on the developed patterns.



FIELD USE

Designed around the rockhounding workflow

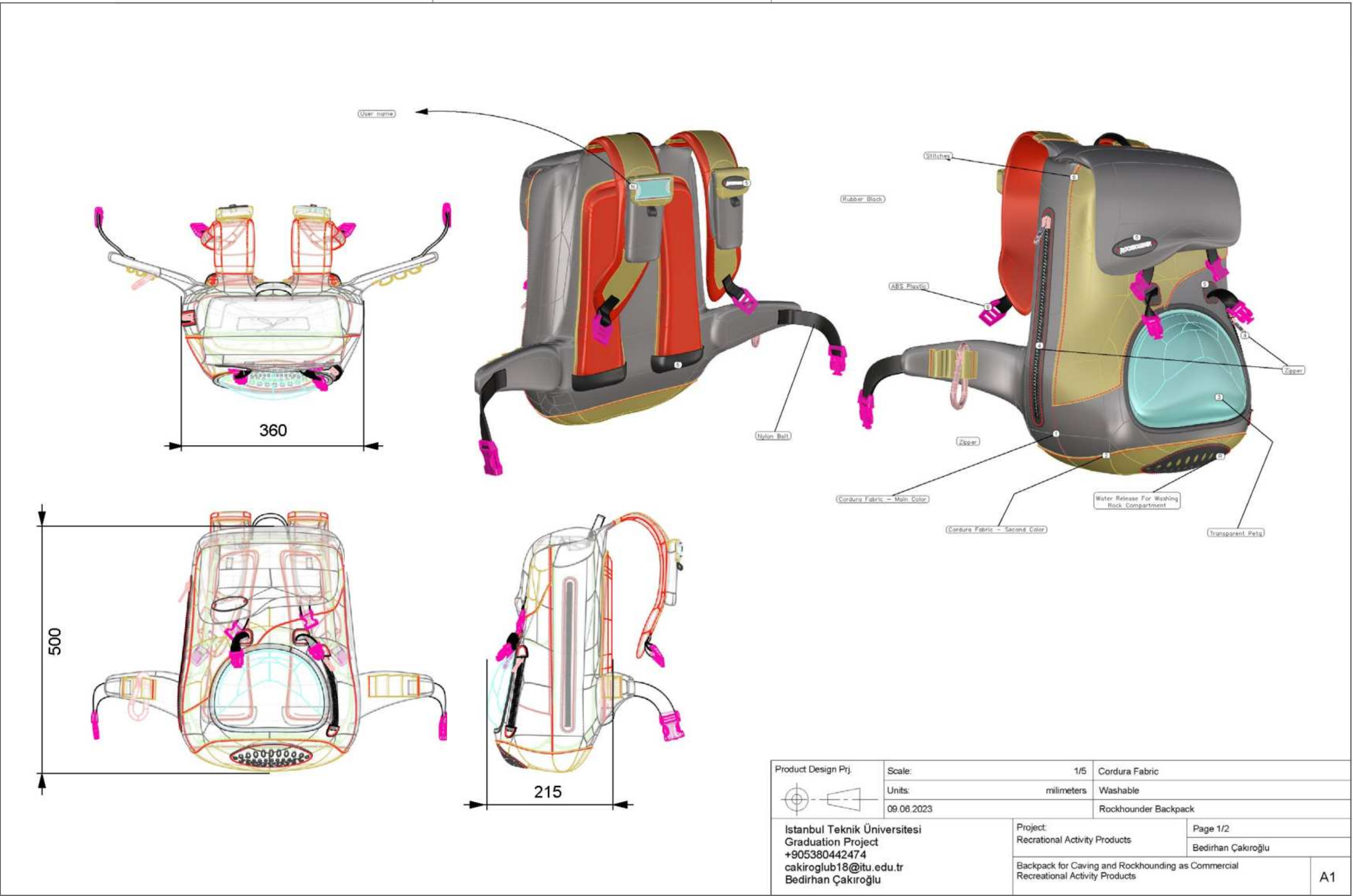
From carrying equipment to collecting, organizing and cleaning finds in the field.



TECHNICAL DEFINITION

Product Definiton

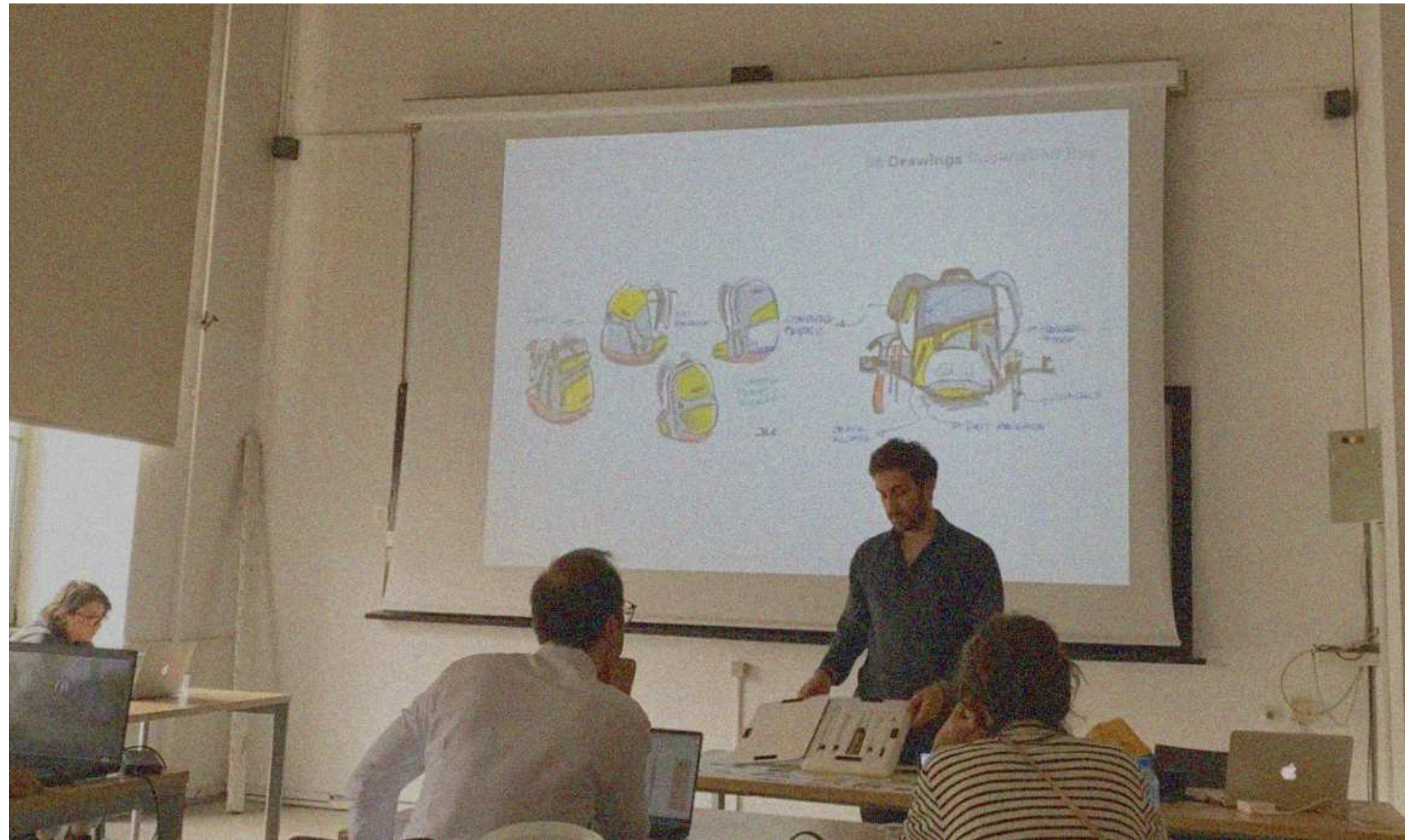
Detailed orthographic views, dimensions and component breakdowns to ensure manufacturability and functionality.



OUTCOME

From Graduation Project to Award-Winning Design

Final outcome of the project



Rockhounder was developed as my graduation project at Istanbul Technical University. Combining field research, product design, prototyping and a supporting digital service.

03

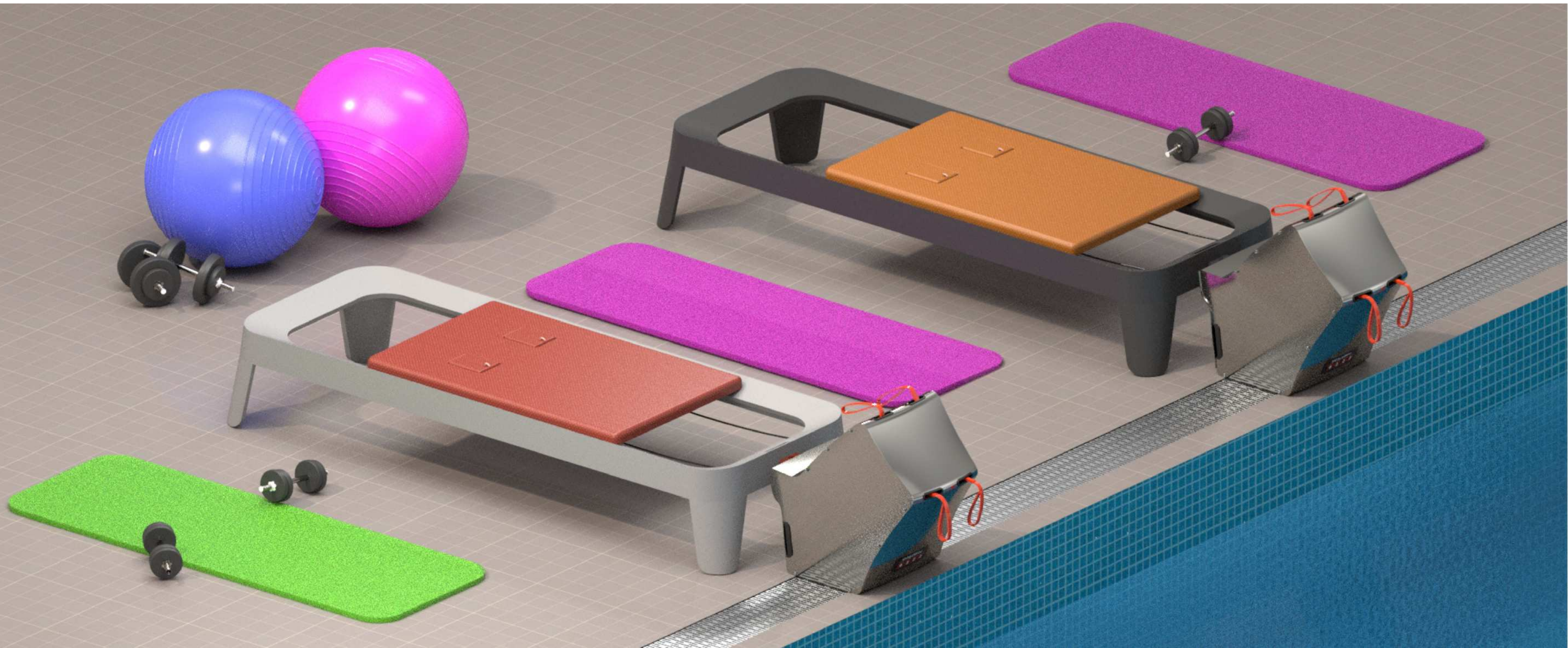
PRODUCT + SERVICE DESIGN

Herptile

Exercise System for Pregnancy

A product-service system designed to support safe and controlled exercise during pregnancy, combining land- and water-based training within an integrated experience.

ITU Industrial Design Dep. · Spring 2022



DESIGN BRIEF

Designing exercise around pregnancy

Understanding the needs, barriers and support system around physical activity during pregnancy.



USER

Pregnant women looking for a safe and accessible way to stay physically active during pregnancy.

KEY NEEDS

Maintain an active and healthy pregnancy

Support the health of mother and baby

Improve confidence and sense of self

PAIN POINTS

Anxiety and fear

Transportation

Cost

Access to appropriate exercise

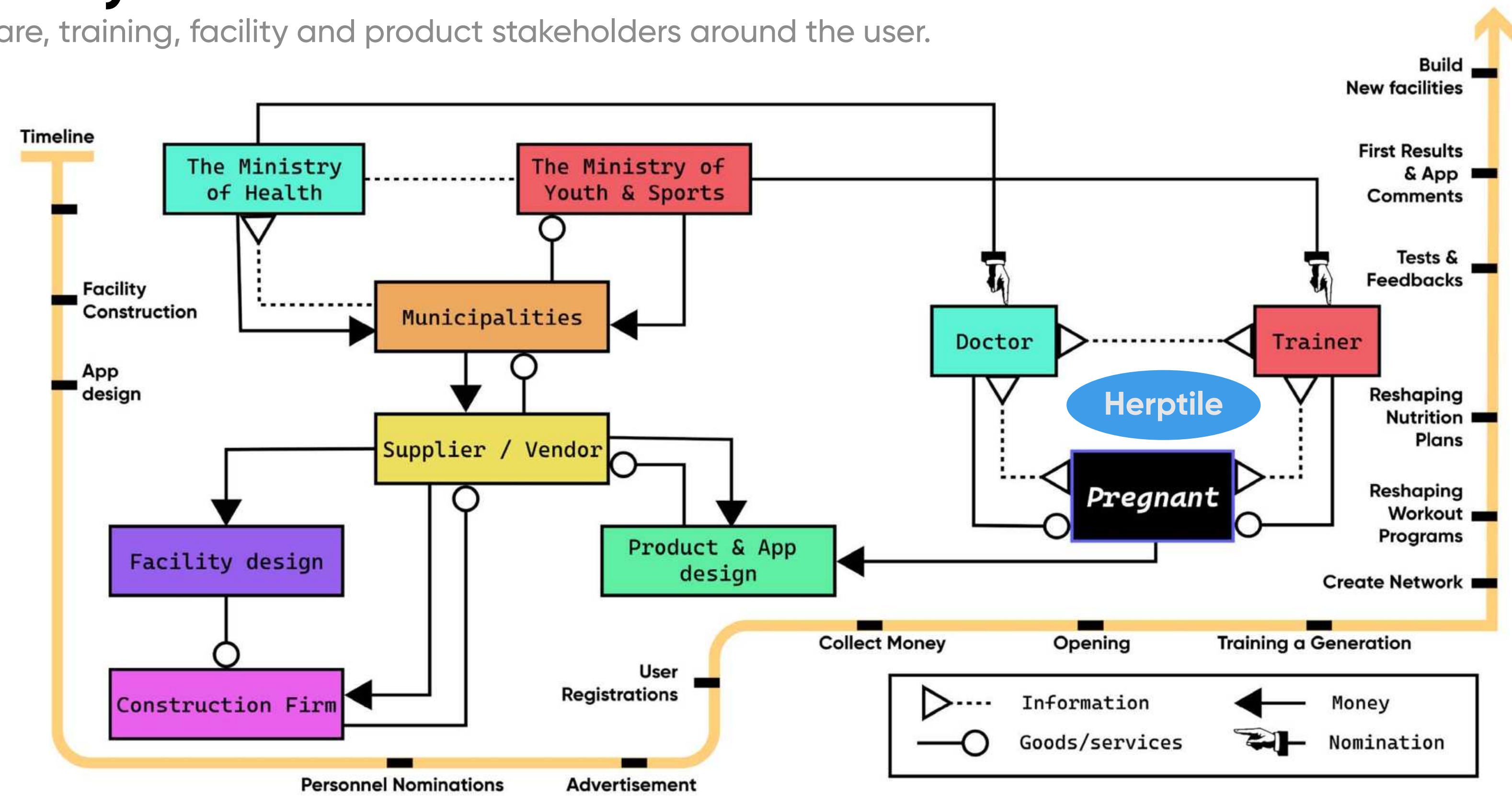
DESIGN OPPORTUNITY

How might we make exercise during pregnancy **safer, more accessible** and **easier** to integrate into everyday life?

SYSTEM DESIGN

How does the system work?

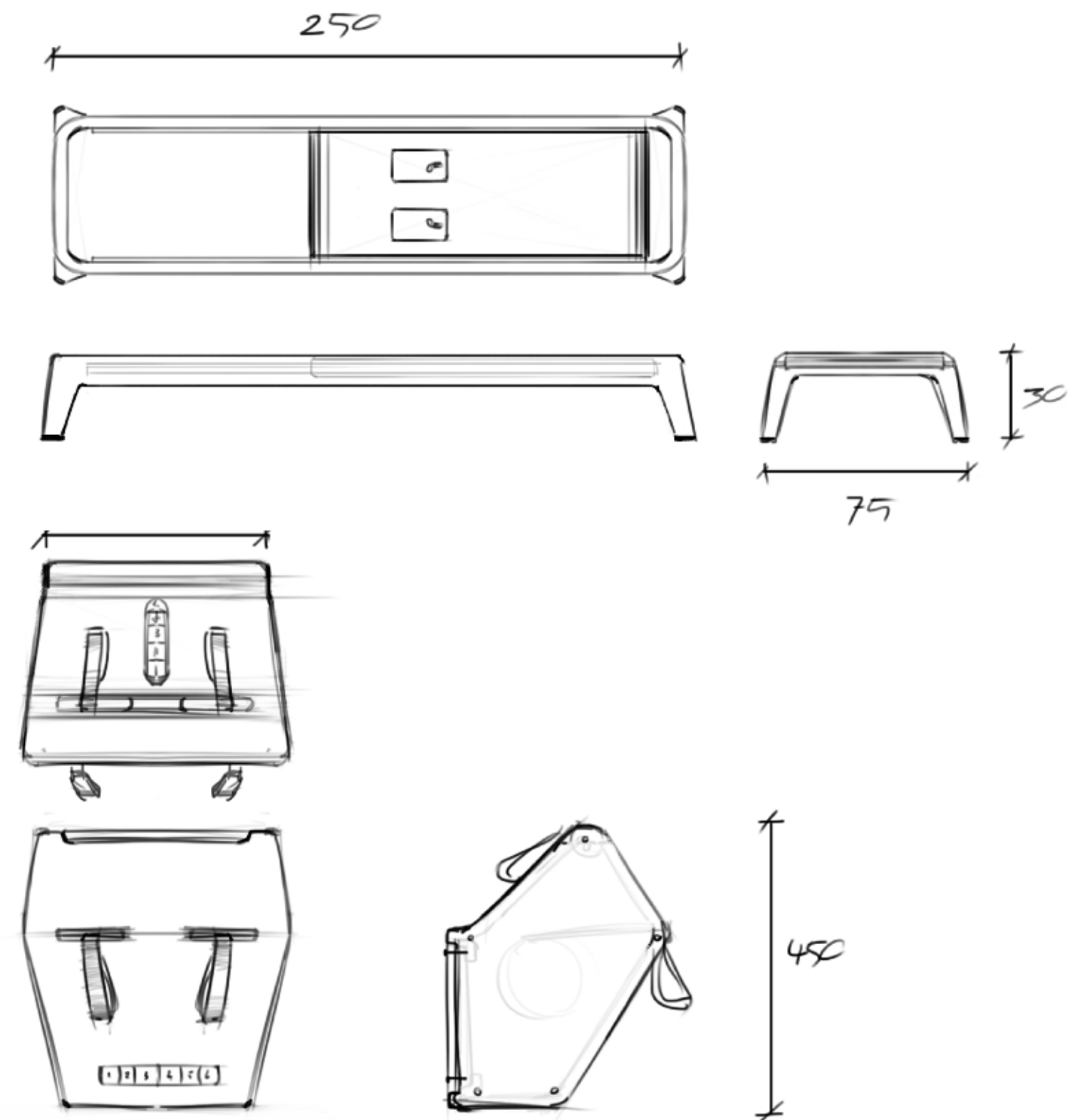
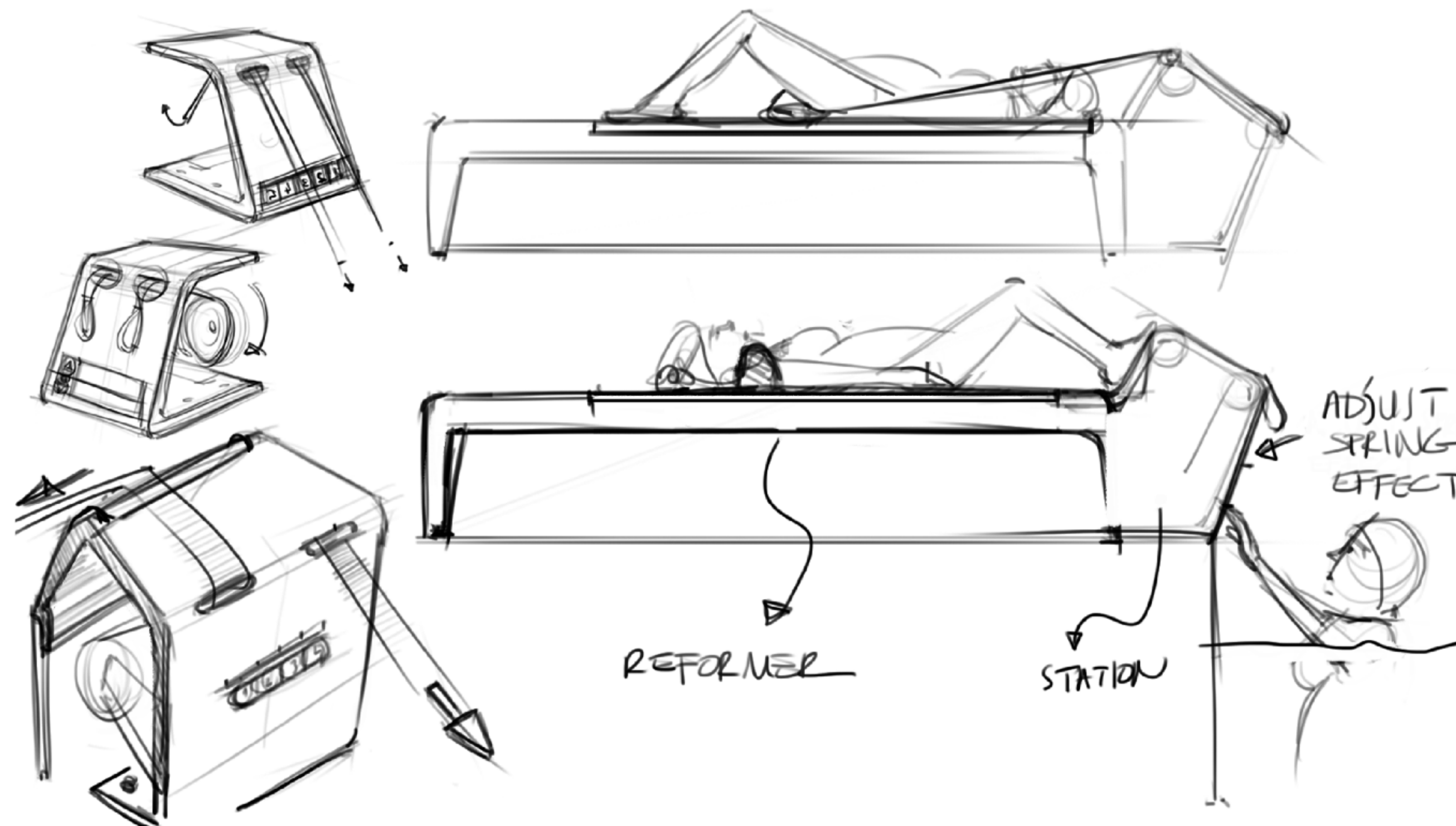
Connecting healthcare, training, facility and product stakeholders around the user.



IDEATION & DEVELOPMENT

Developing the station and reformer

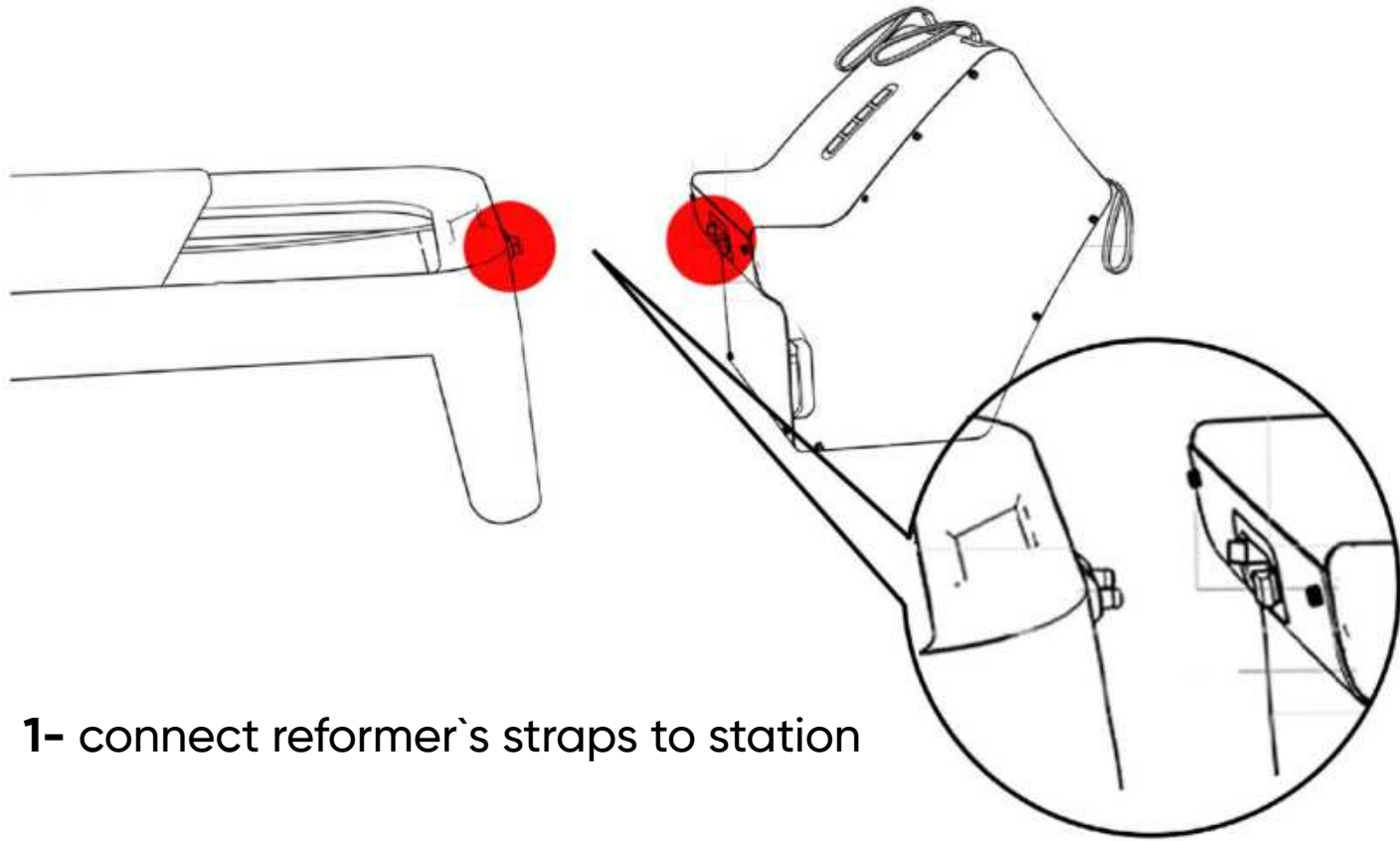
Exploring the interaction, proportions and mechanical relationship between the two product components.



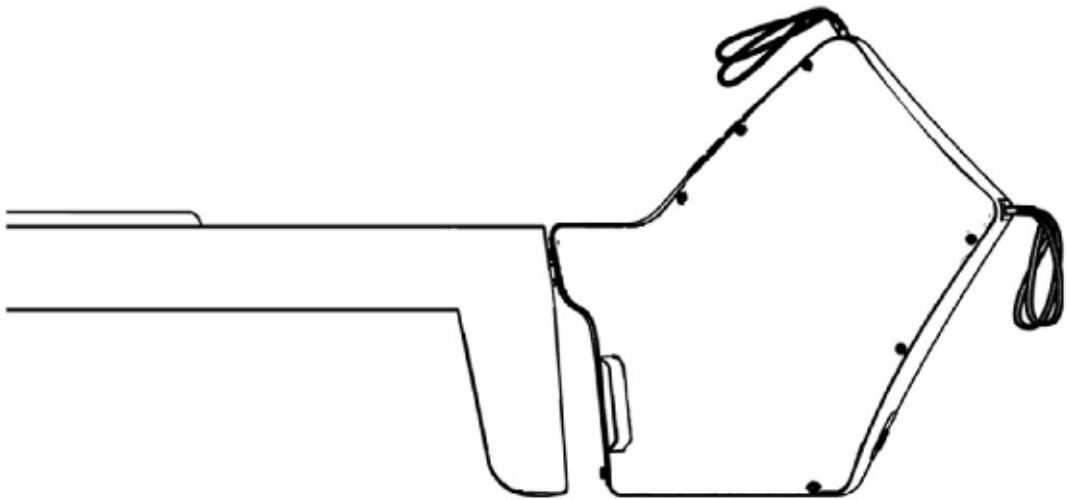
USE SCENARIO

Setting up the system

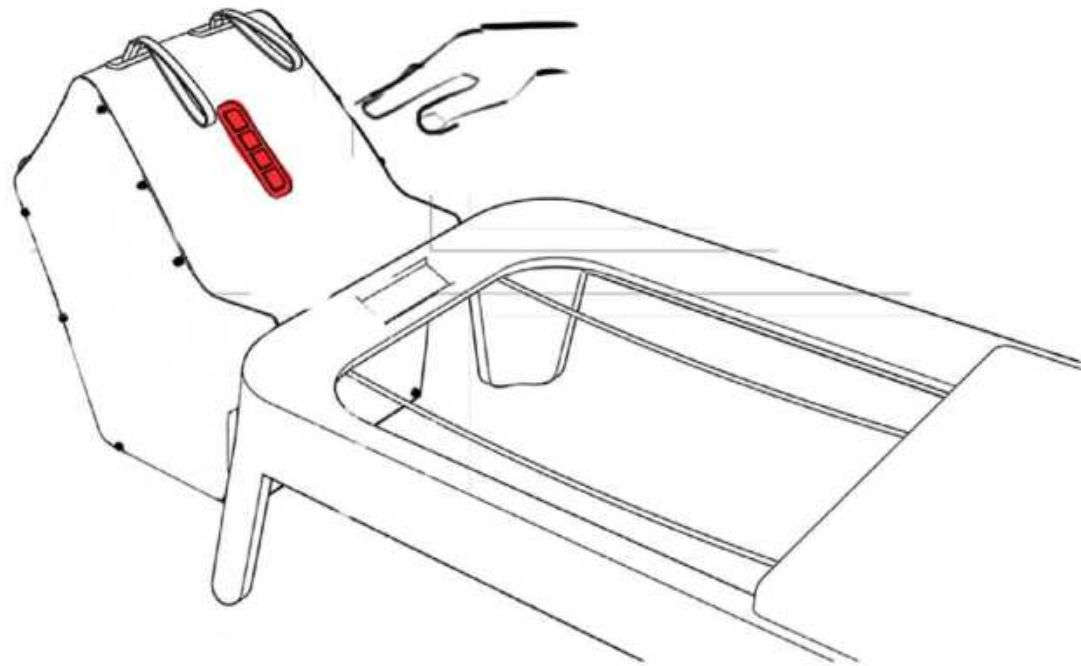
Connecting the reformer to the station and adjusting resistance for land- and water-based exercise.



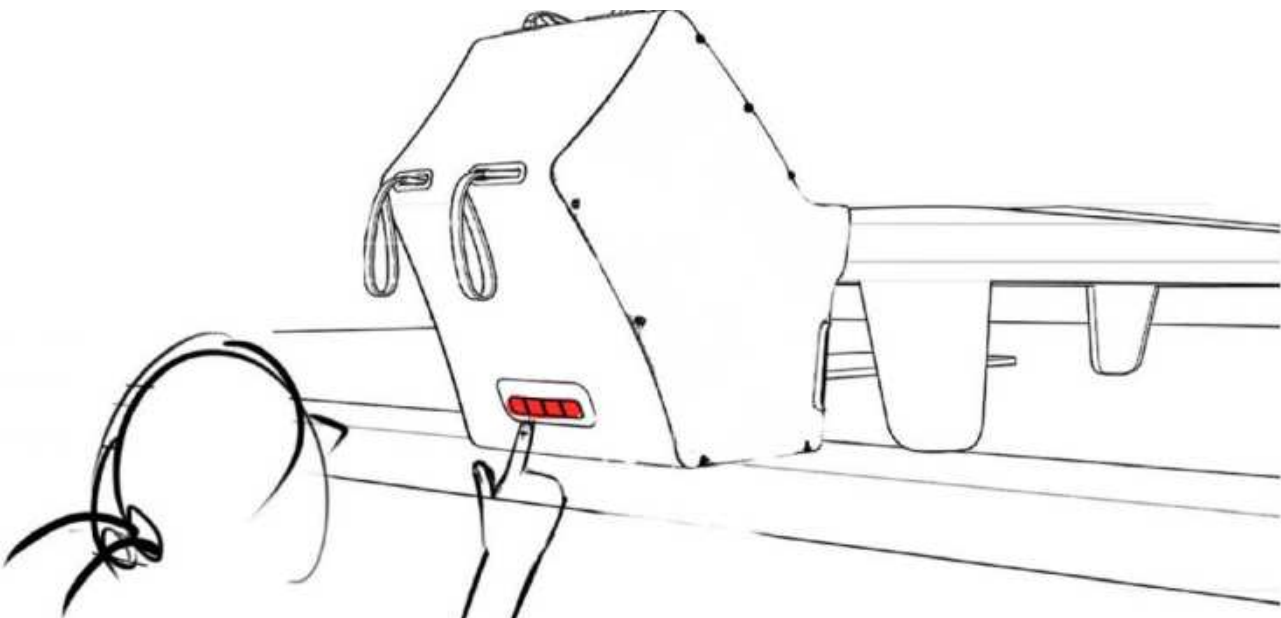
1- connect reformer's straps to station



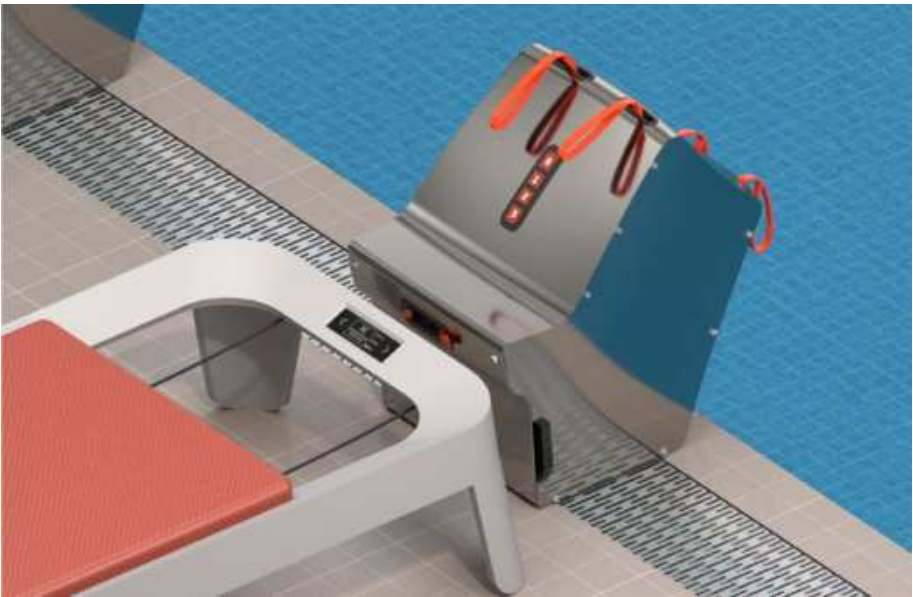
2- reformer and station are connected



3- select spring stiffness



4- spring stiffness for users in water



5- trainer starts reformer



6- reformer screen

PRODUCT DESIGN

The Reformer

A compact exercise platform designed for controlled movement and adjustable resistance.



PRODUCT DESIGN

The Station

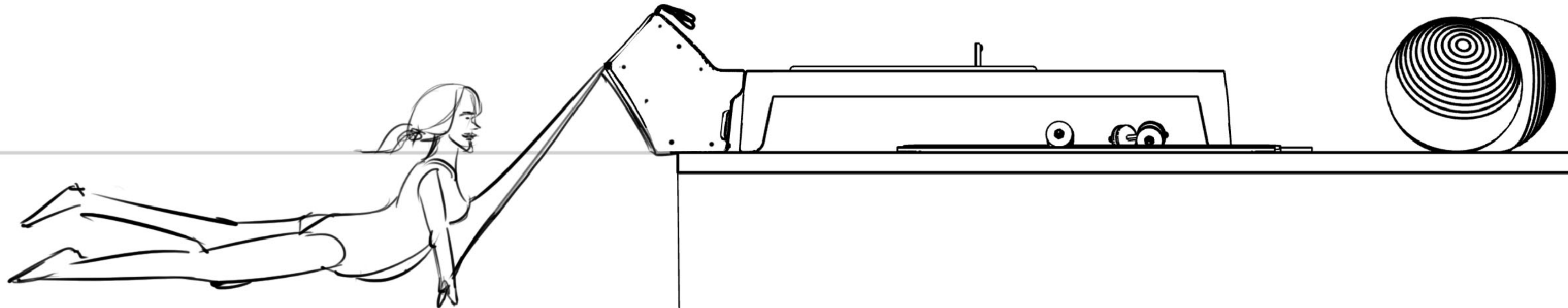
A poolside interface connecting the reformer with the water-based exercise experience.



TECHNICAL DEFINITION

Product Definition

Integrating the reformer and poolside station into a controlled Pilates experience.



Herptile brings together product and service design to create an integrated exercise experience for pregnant women, supporting controlled Pilates training both on land and in water.



Robotistan Product Nr.: 18603

PRODUCT + MANUFACTURING DESIGN

Peanut – Mini Tracked Robot

Industrial Design & Design for Assembly

An Arduino-based mini tracked robot redesigned as both a physical product and an assembly experience, developed for commercial production at Robotistan.

Robotistan Product Development Internship_ 2022
Duration 4 weeks
Role Product Design_ Mechanical Design_ Assembly Documentation

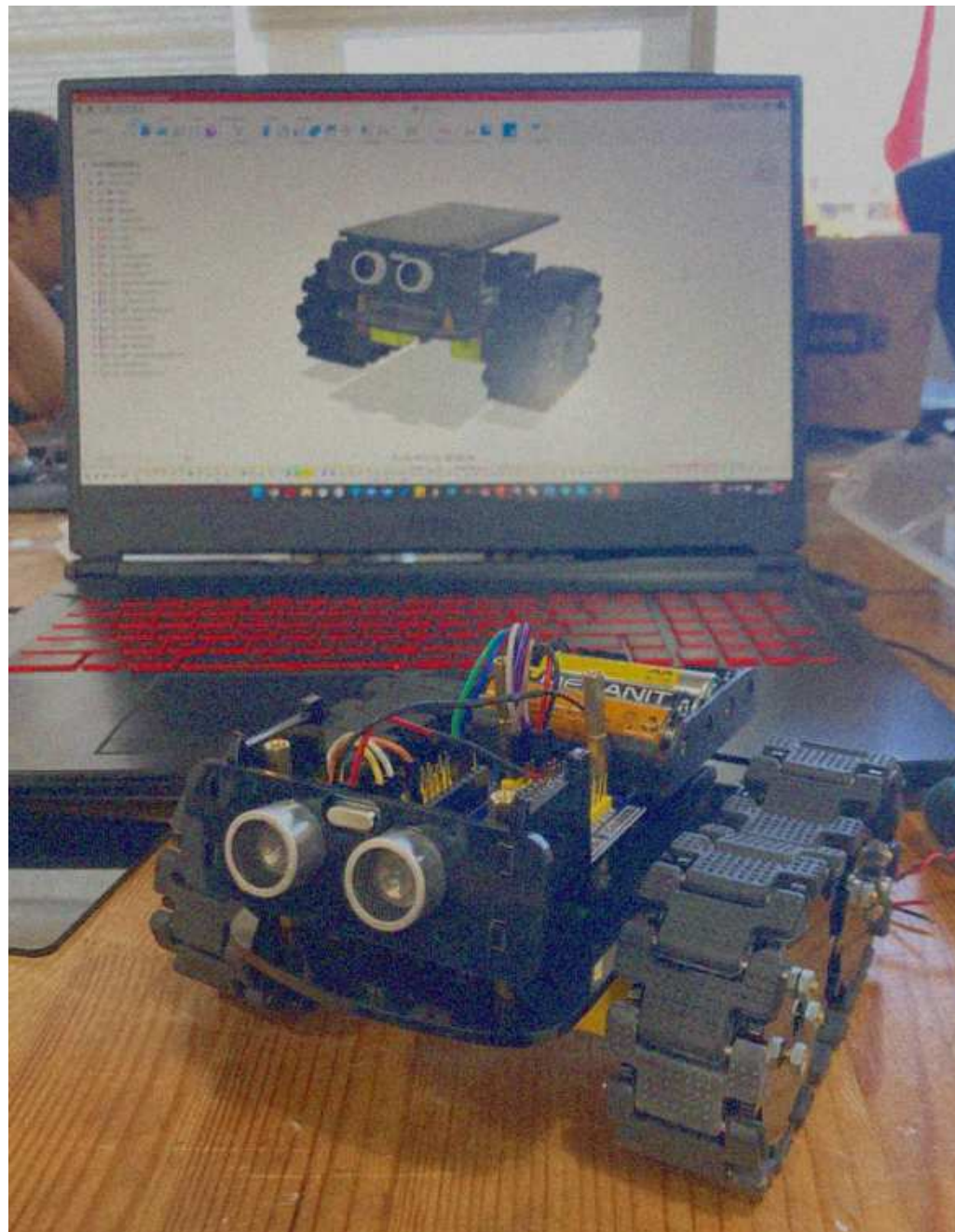
04



DESIGN BRIEF

Designing the kit, not just the robot

With an educational robotics kit, the hard part is usually not the electronics. It is letting the user put the parts together in the right order and the right orientation.



Peanut had to work as two things at once, a piece of hardware and an assembly experience. The electronics are standard, so the real design problem was a buildable, legible, repeatable kit.

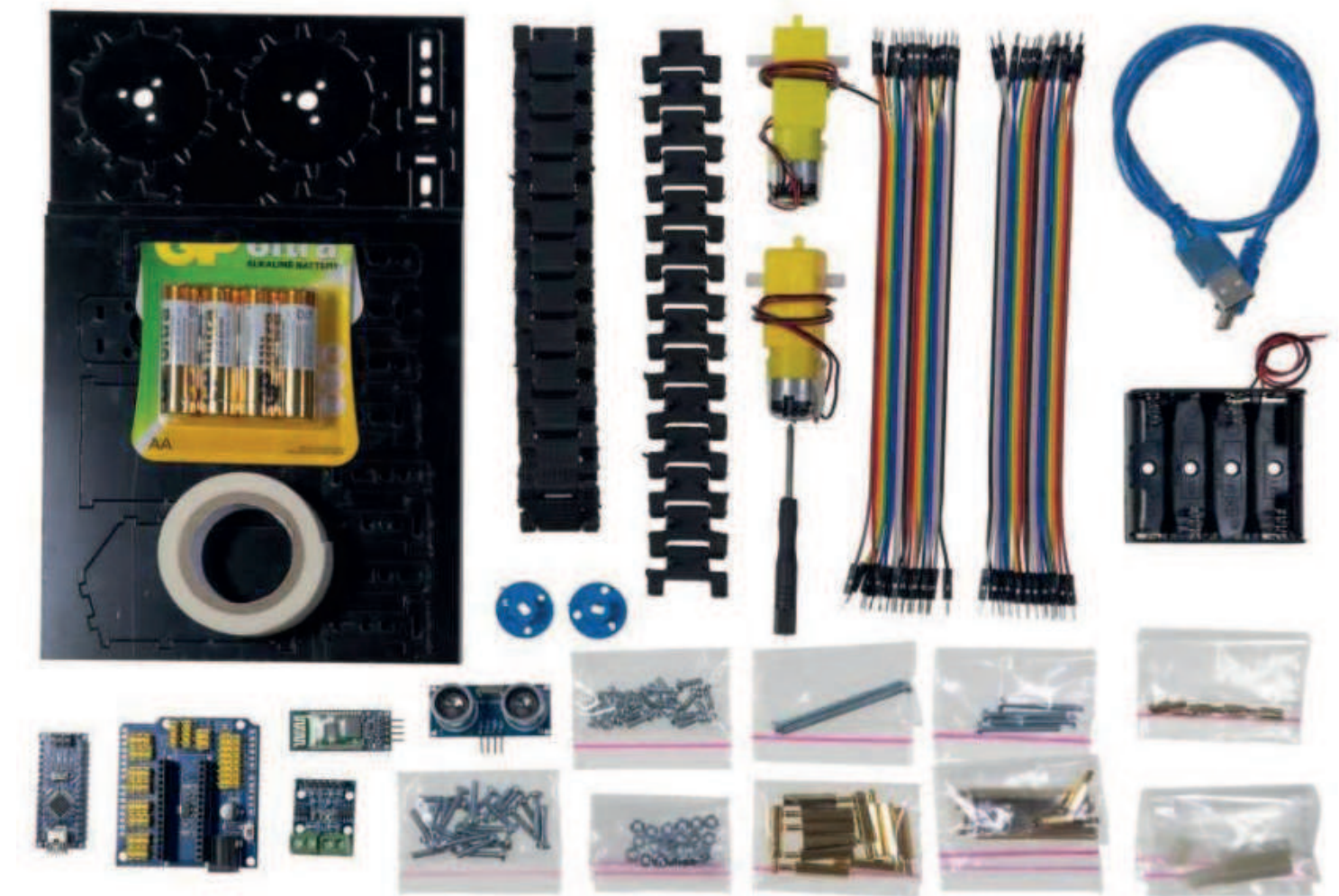


Image Reference: <https://www.robotistan.com/mini-paletli-robot-platformu-elektronikli>

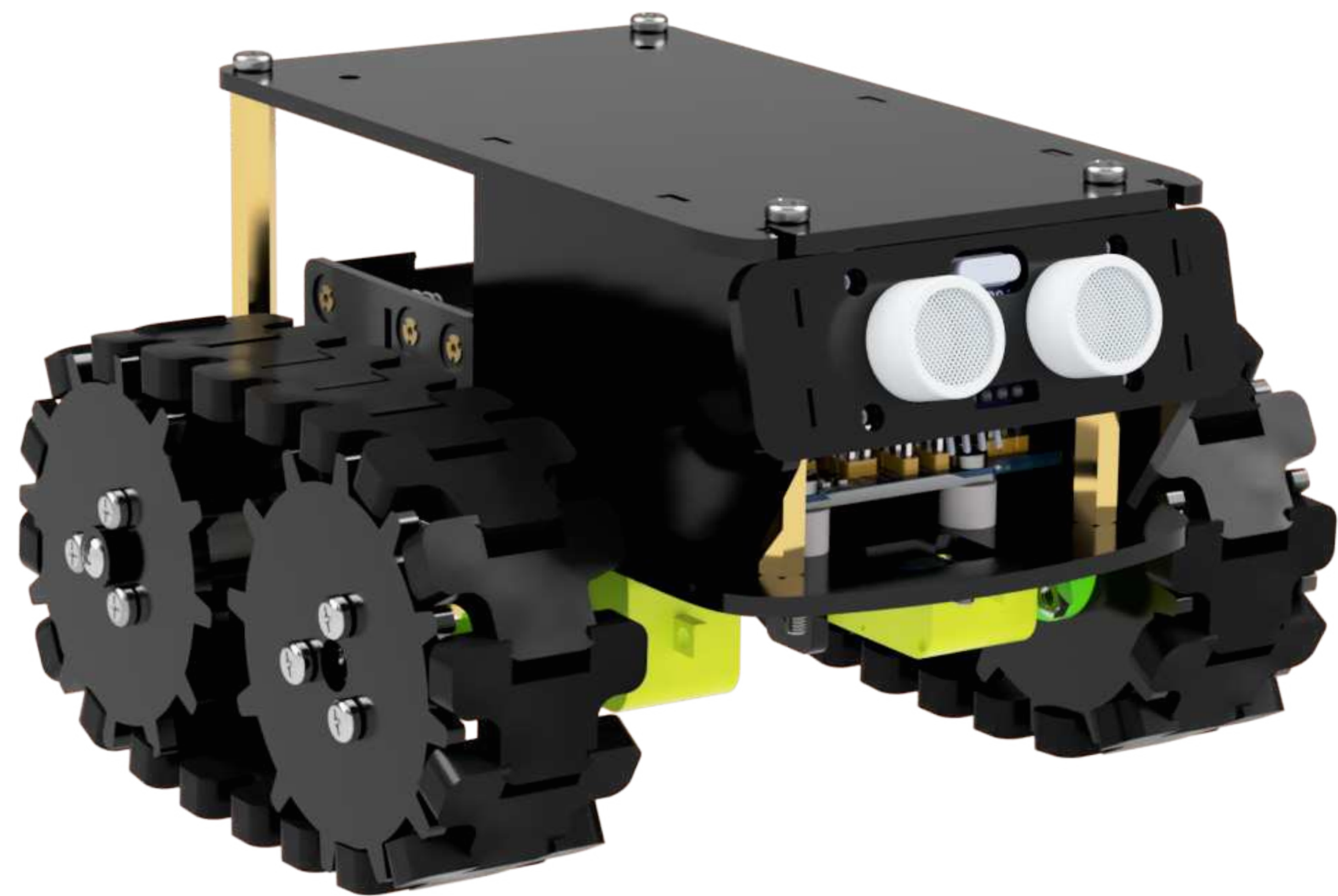
- All-Terrain Mobility_** Tracked drive enables movement across uneven surfaces.
- Maneuverable Platform_** Differential tracked drive enables agile movement and turning.
- Integrated Systems_** Supports sensors, actuators and additional robotic modules.
- Autonomous Navigation_** Can be developed into an autonomous mobile robot.

DESIGN CRITERIA

Deciding what “right” meant

Four requirements framed every decision.

CRITERION	REQUIREMENT
Assembly ease	No electronics experience required
Electronics integration	Arduino + Bluetooth + ultrasonic sensing
Manufacturability	Laser-cut acrylic construction
Cost	Educational-kit price range



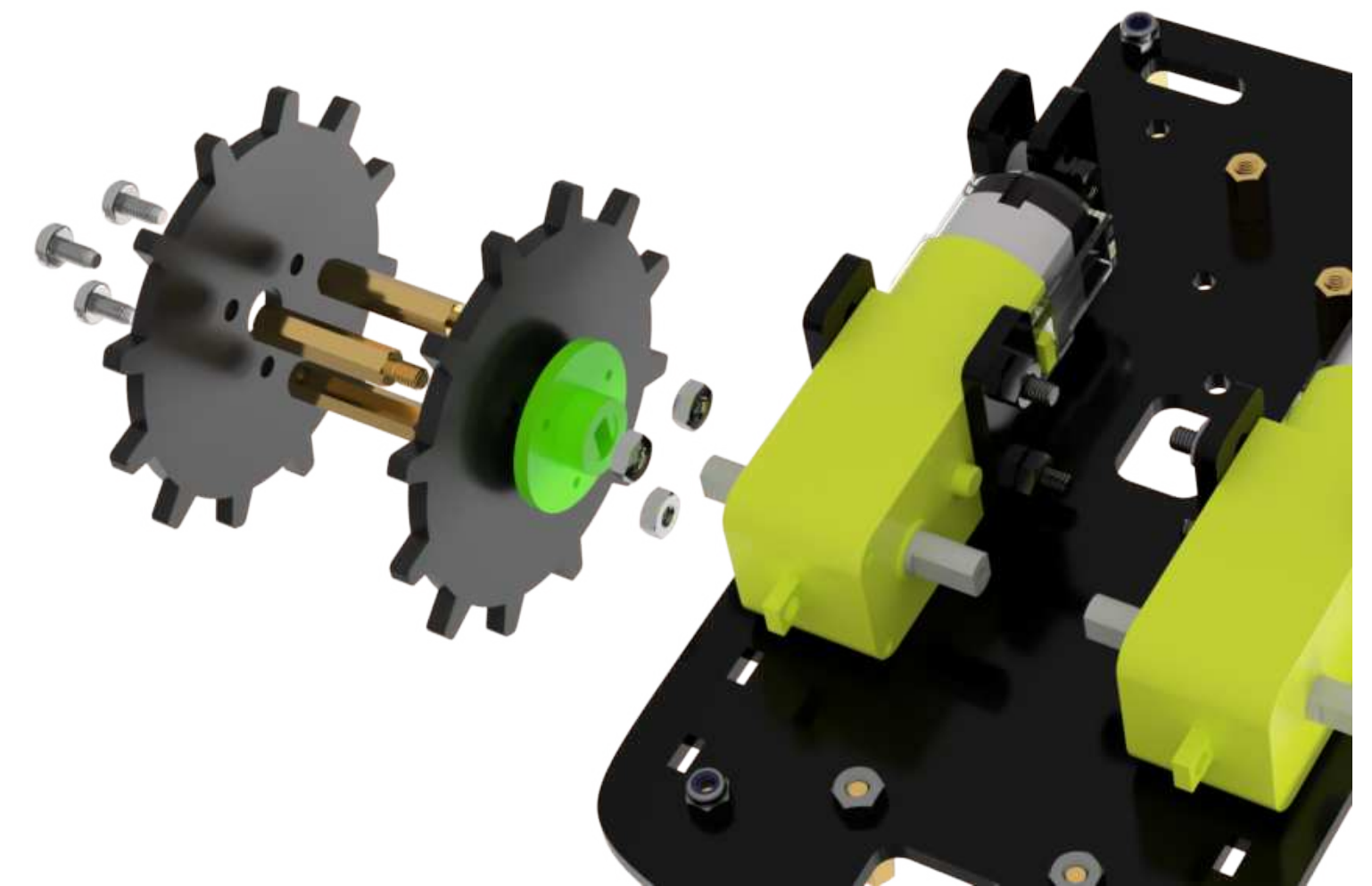
CONCEPT & ARCHITECTURE

A layered body on brass standoffs

The body is built from stacked laser-cut acrylic plates joined by brass standoffs, simplifying production, component organization and assembly.



Different standoff heights organize the electronics between layers, integrating the Arduino Nano, Bluetooth module, ultrasonic sensors, geared DC motors and battery holder.



Multiple CAD iterations were used to refine component layout, body proportions and drivetrain

DESIGN FOR ASSEMBLY

The assembly guide is part of the design

The clearest design-for-assembly proof is the step-by-step visual manual.

AVAILABLE ONLINE

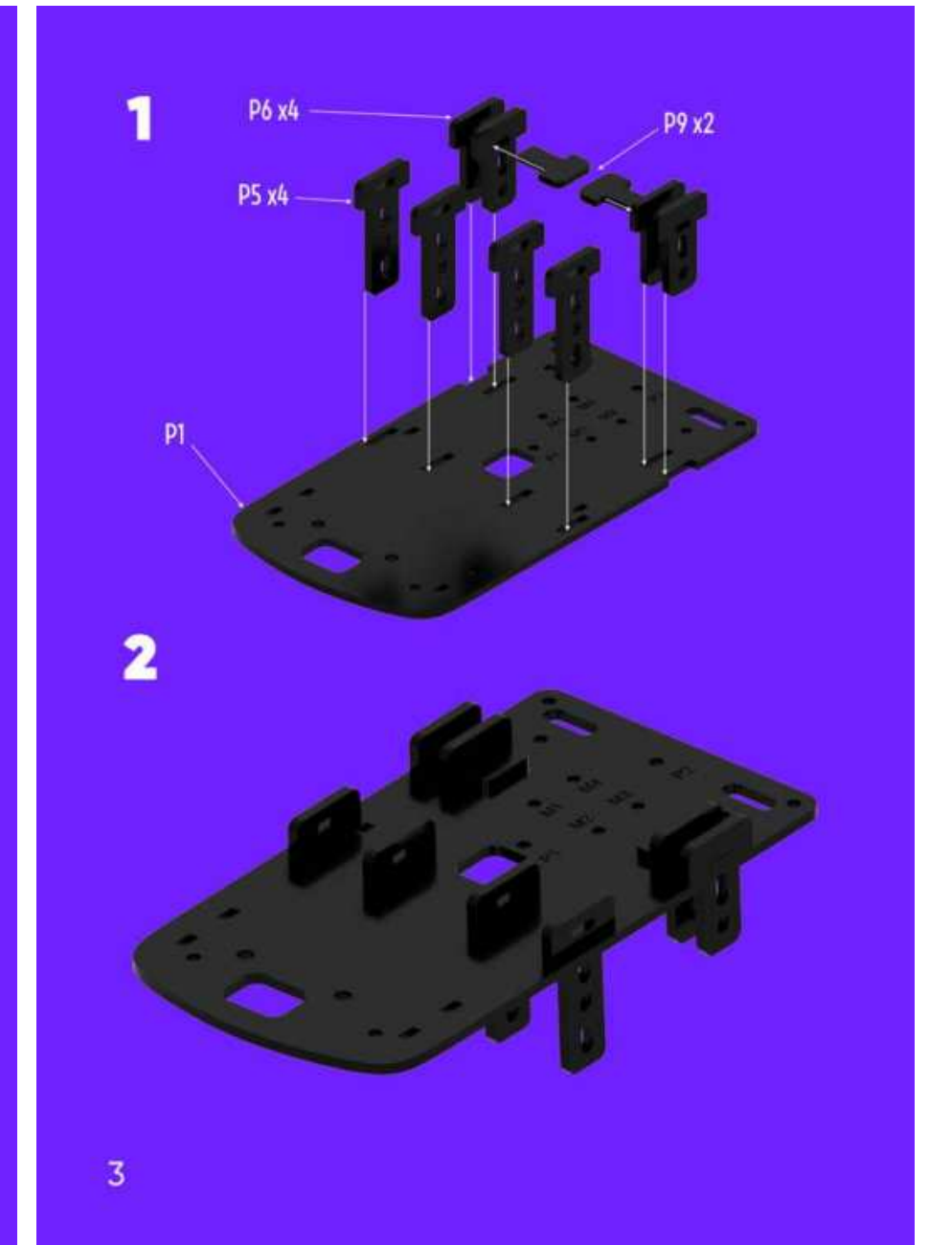
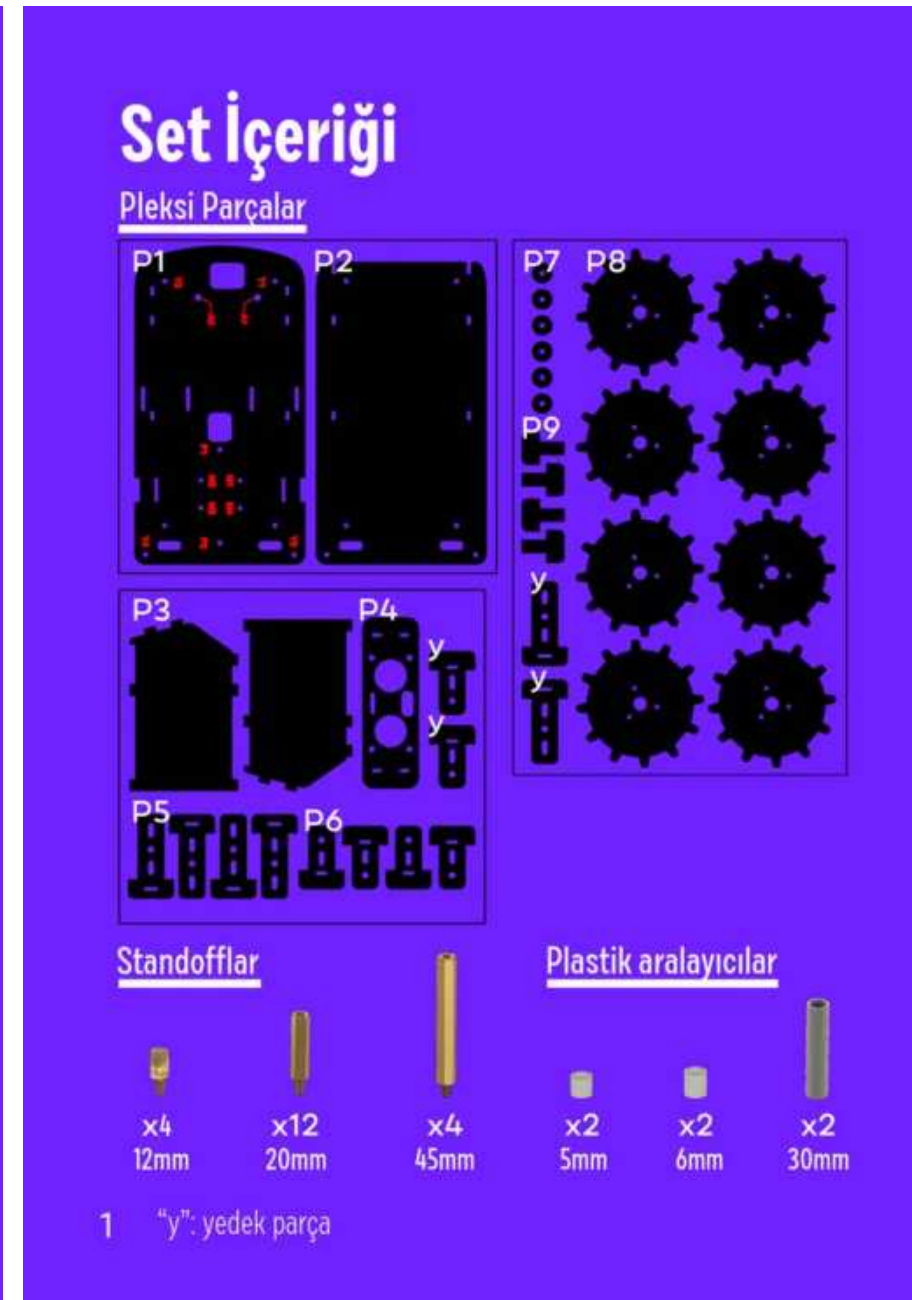
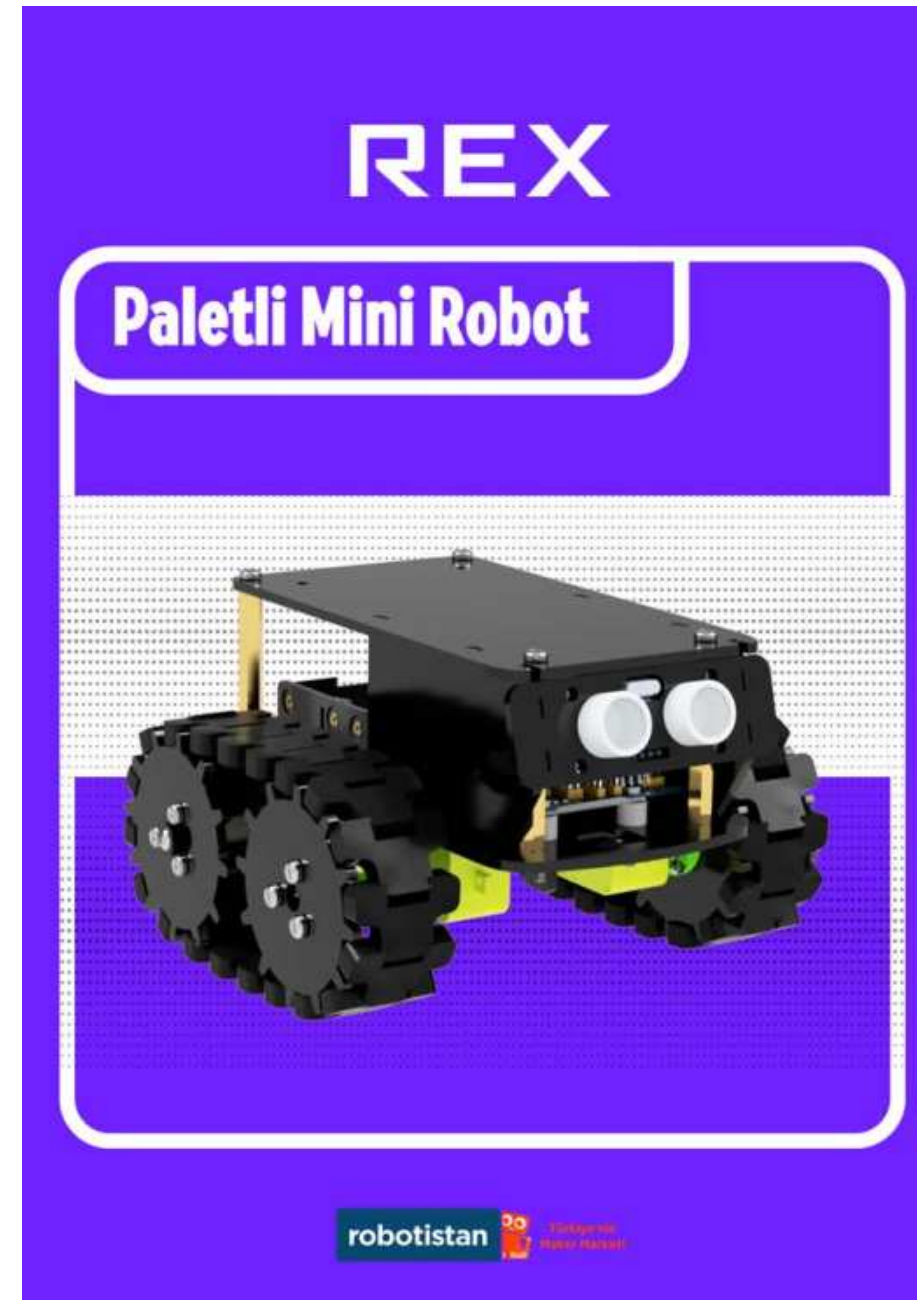
Available for purchase through the Robotistan website.

REX – Mini Tracked Robot

Commercially launched by Robotistan as Peanut.

COMMERCIAL PRODUCT

Peanut was launched as a commercial Robotistan product, complete with dedicated packaging and an 18-page assembly manual.



05 Selected Works



CONSTRUCTION DESIGN

TUS Metallbau

.1 Metal construction design,
workflow over +150 Projects.



PRODUCT DESIGN

F1 SIM Steering Wheel

.2 Personal product design project.
Real production.



FURNITURE DESIGN

Coffee Table

.3 Personal product design project.
Real production.



GLASS DESIGN

SPIEGELAU

.4 Glassware concept based on
Spiegelau's design language.

05.1 SELECTED WORKS

TUS Metallbau

Technical Design & Construction

Design, construction and production documentation for metal construction projects, including railings, stairs, canopies and custom structures.

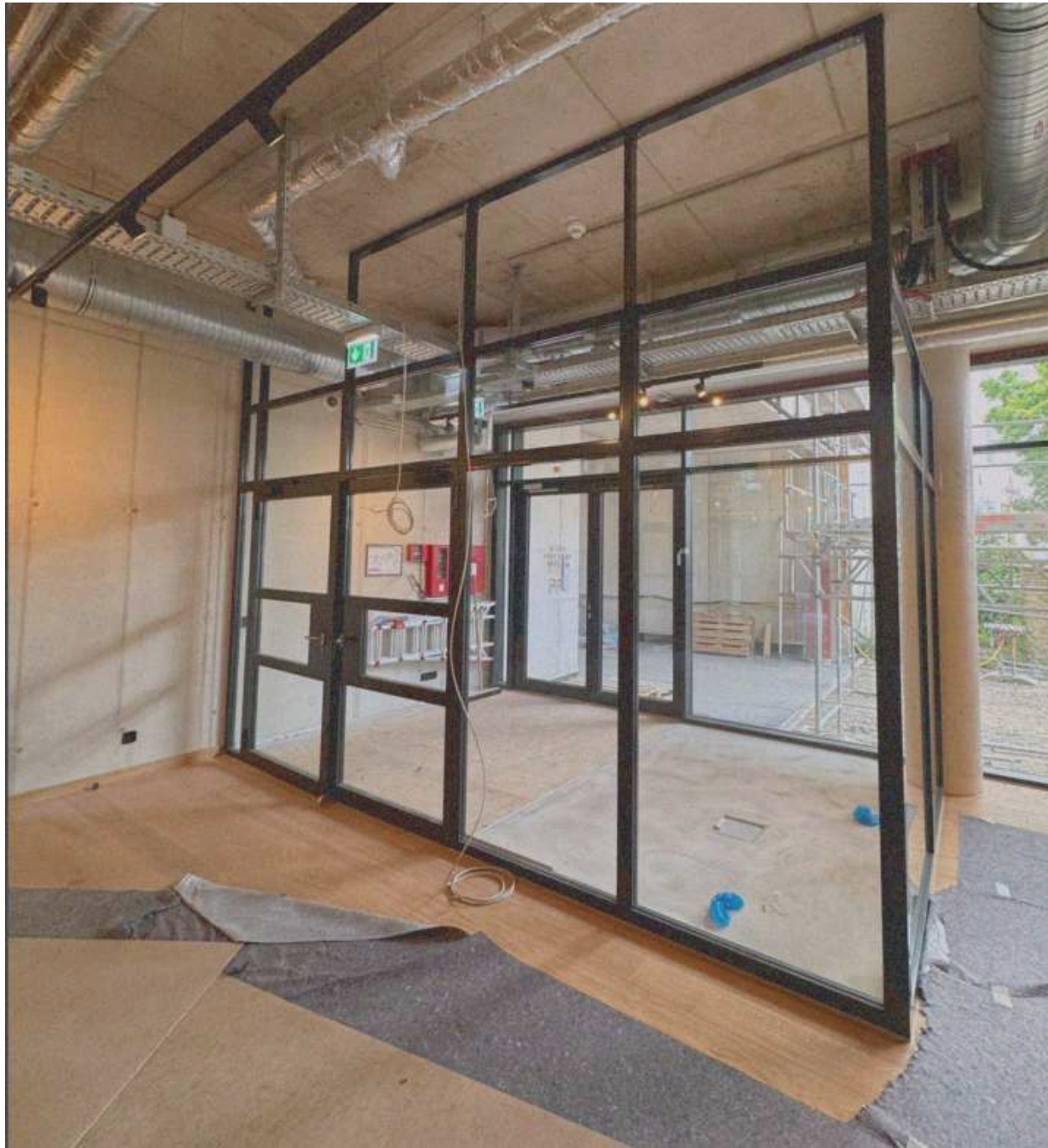
3D modeling & technical drawings

Production documentation

Material planning

Site measurements

Coordination with production and assembly teams



STAYERY HOTEL - OSNABRUCK



PRIVATE CUSTOMER - PADERBORN

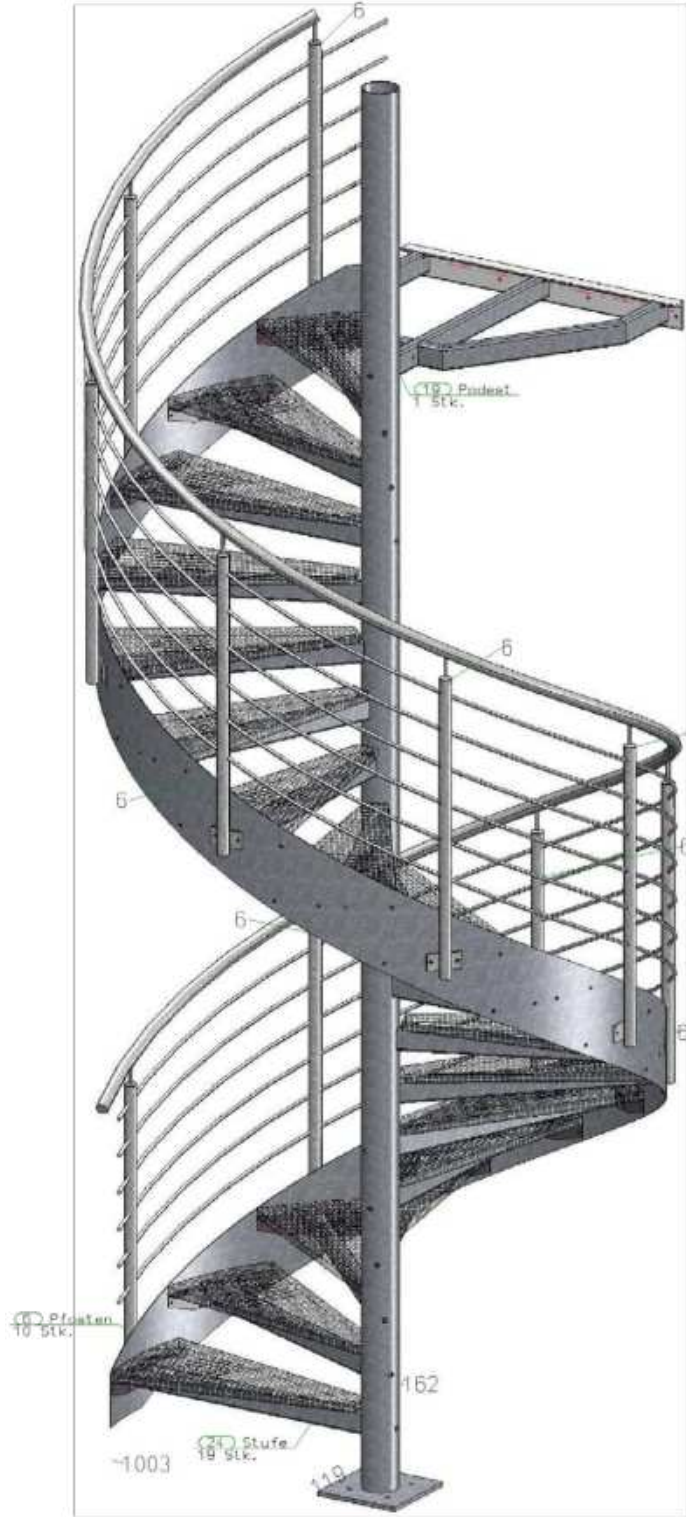
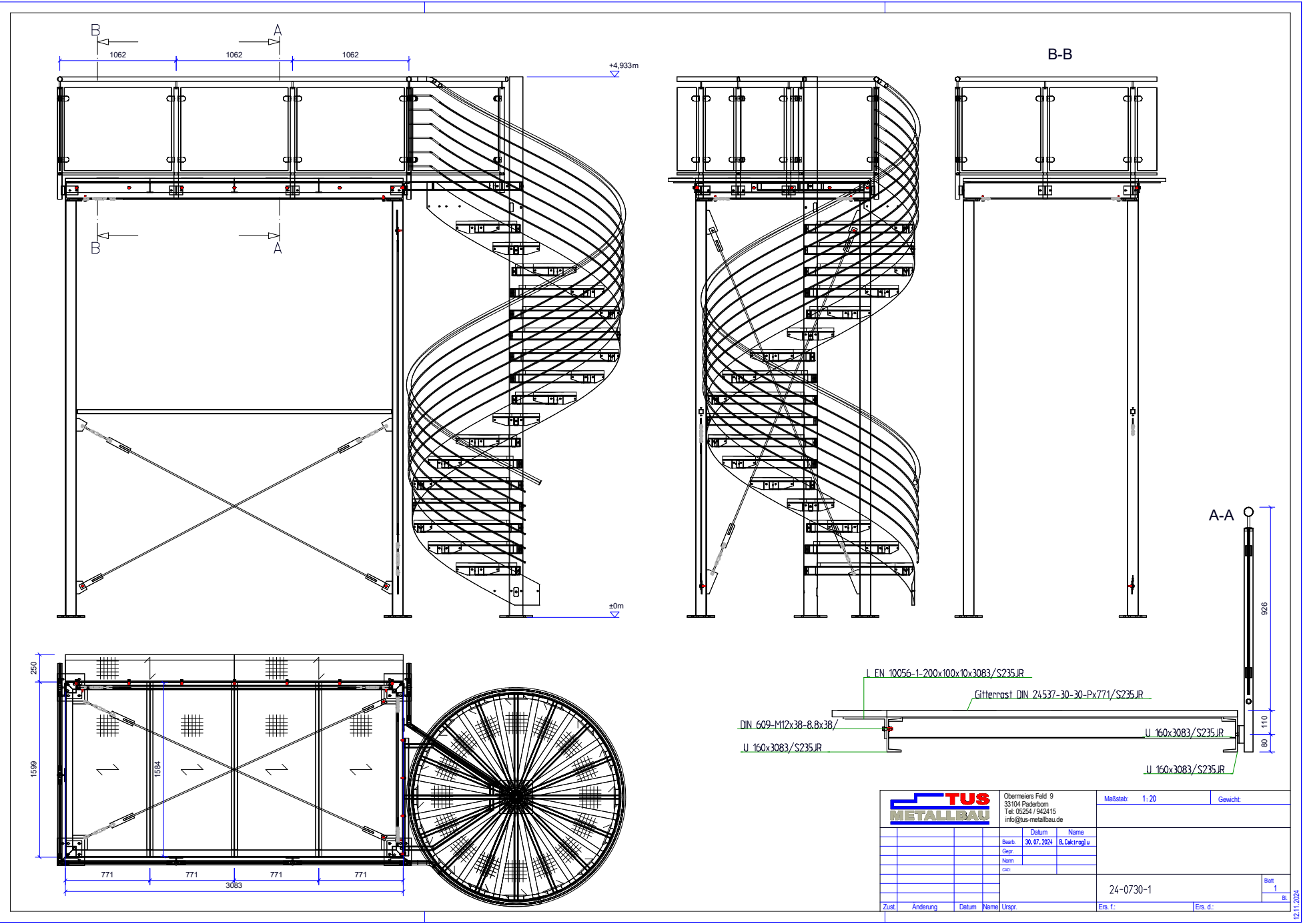


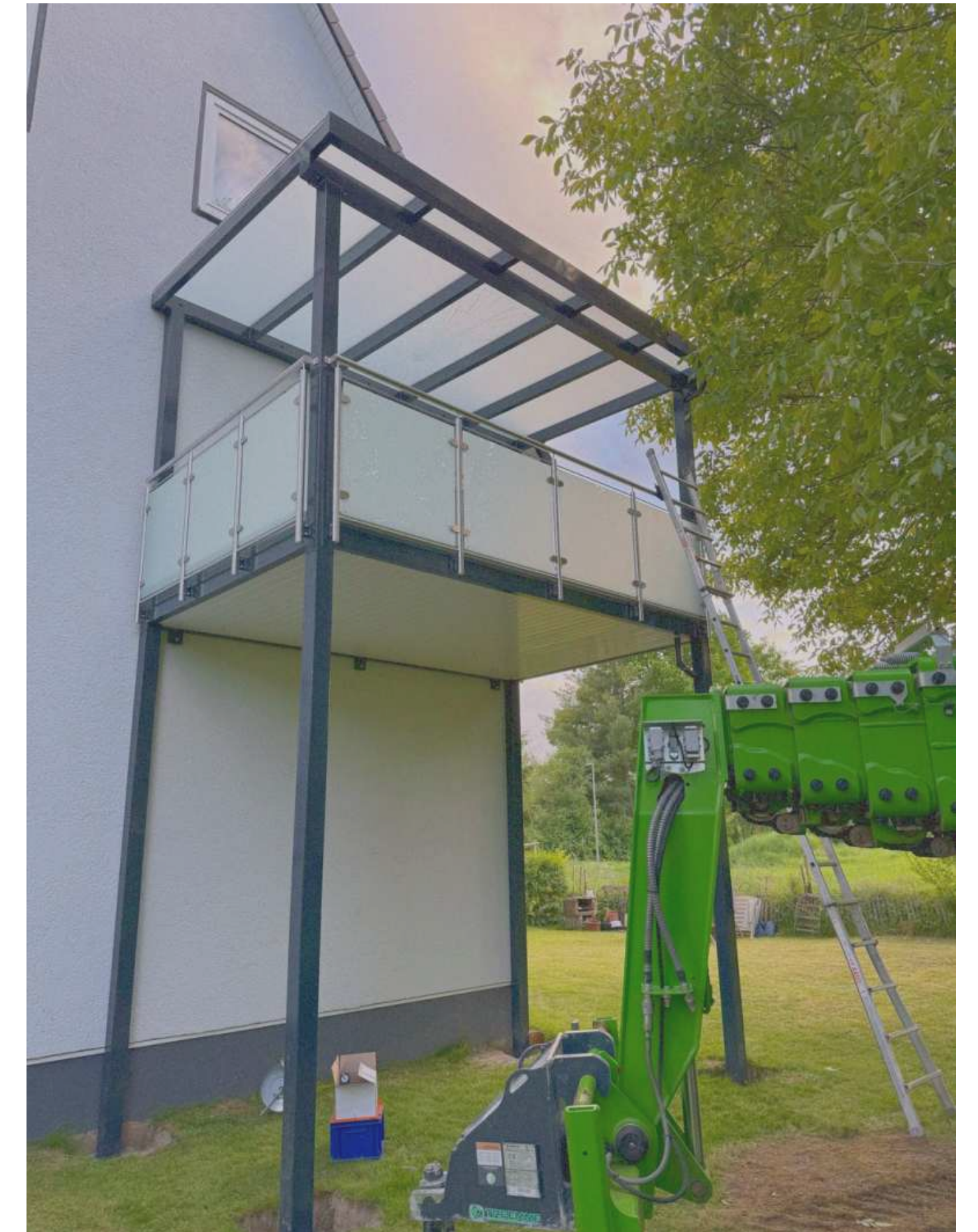
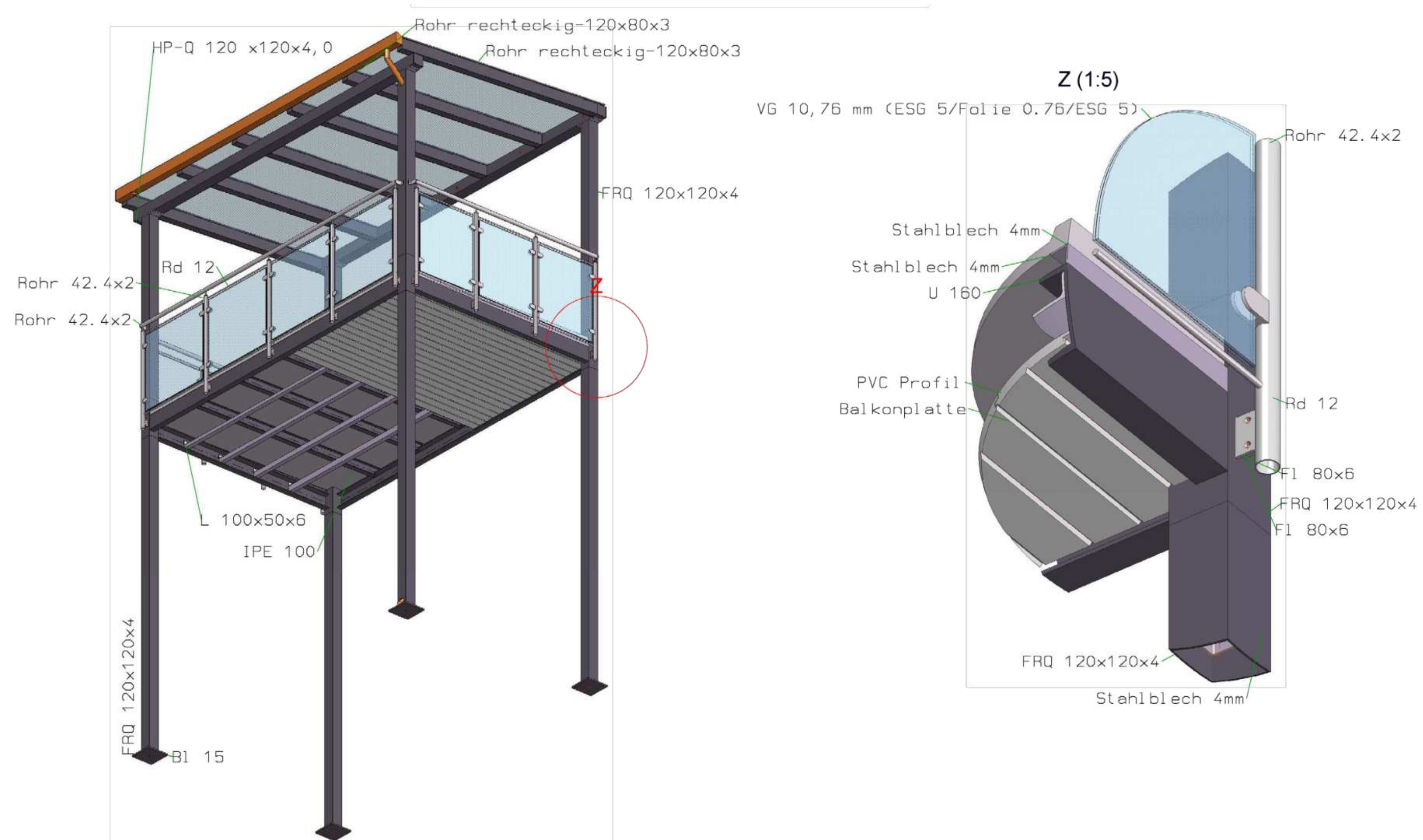
CARL ZEISS AG - BRAUNSCHWEIG

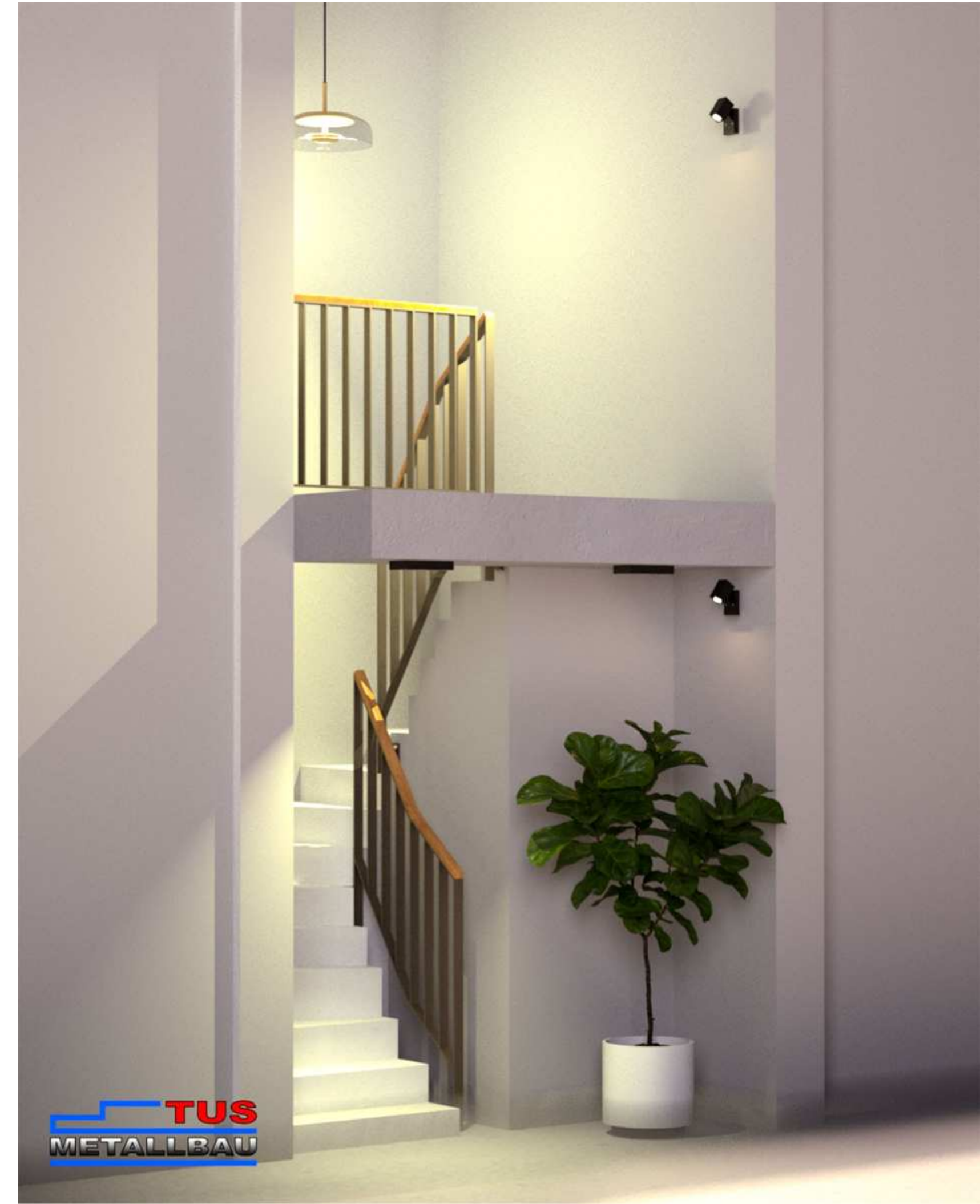
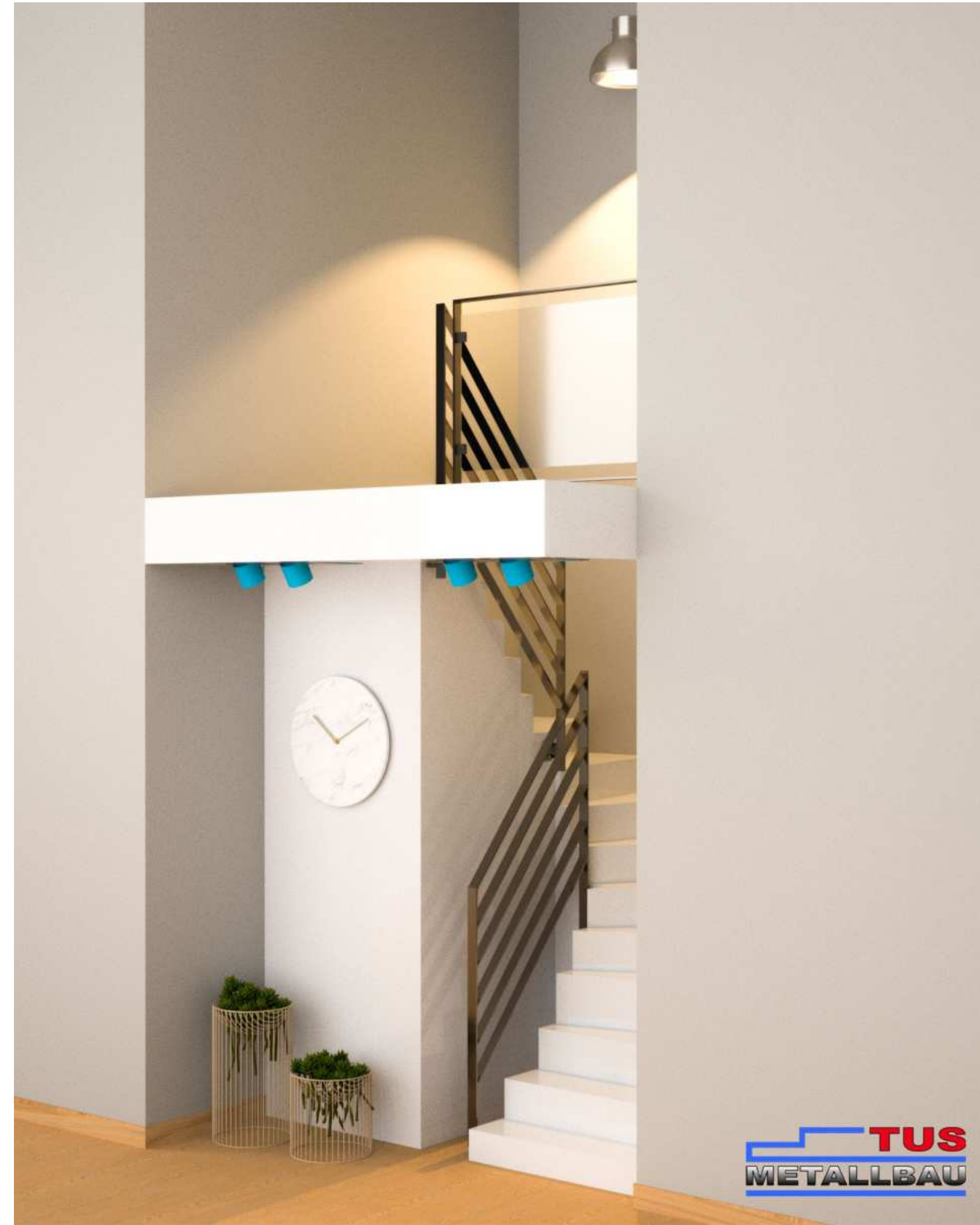
05.1 SELECTED WORKS

From measurement to production

Site measurements, detailed CAD models and production-ready drawings developed in coordination with manufacturing and assembly teams.







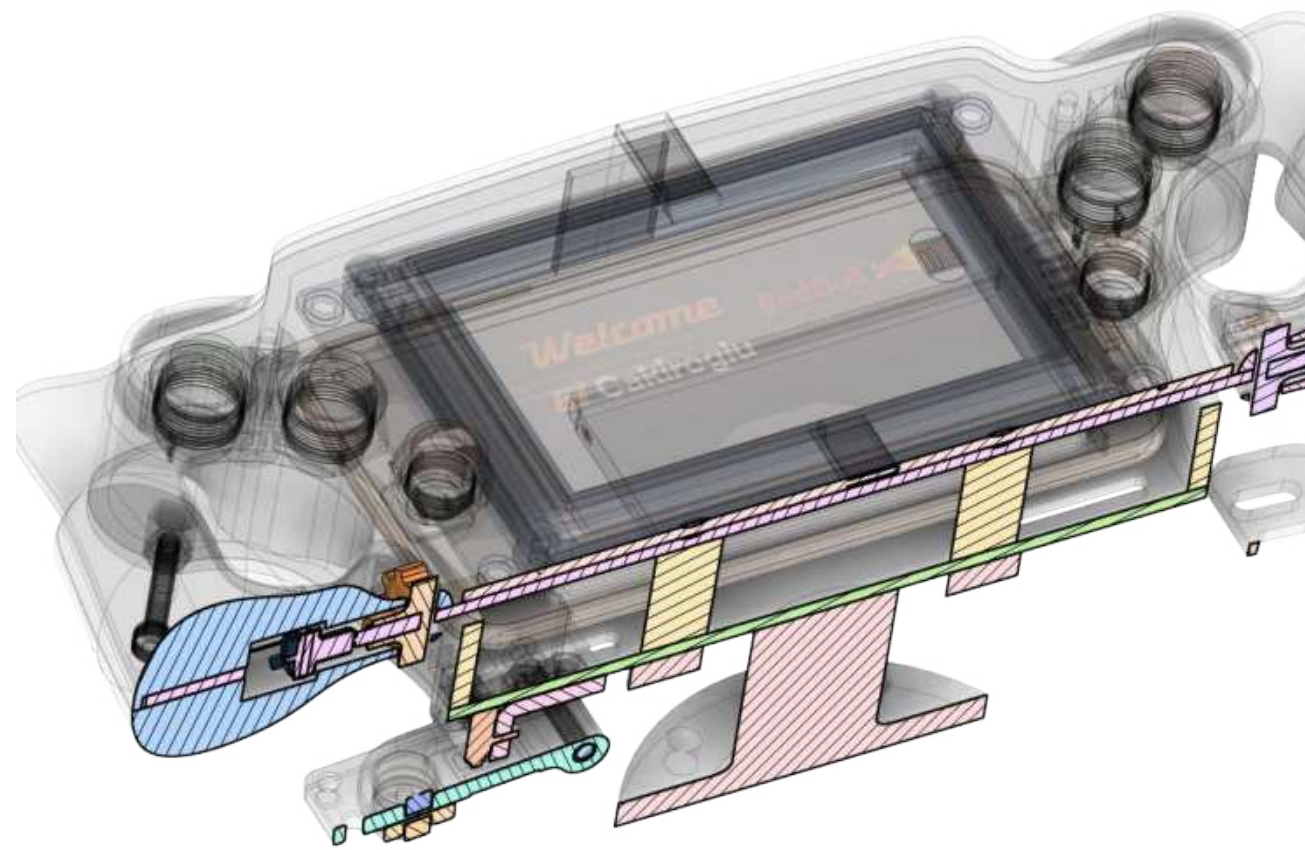
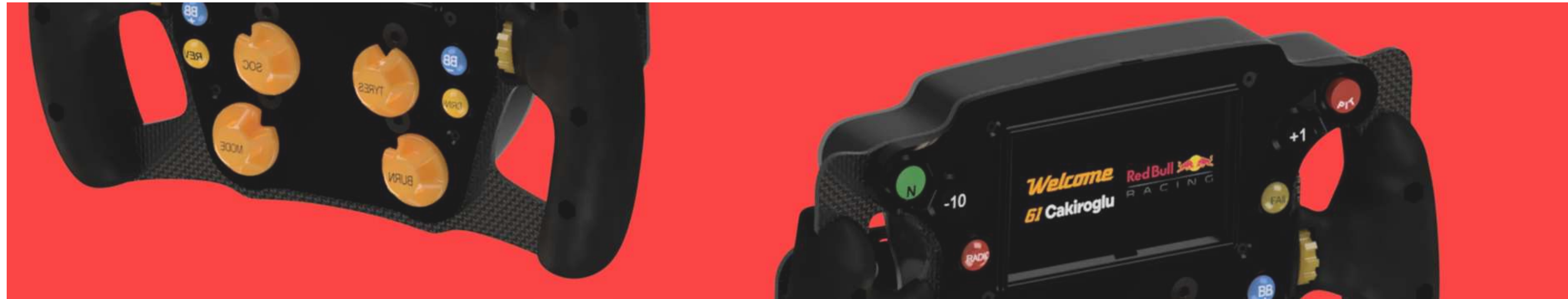
05.2 SELECTED WORKS

F1 SIM Steering Wheel

Electronic Product _ Designed & Built

Arduino-Based F1 SIM Steering Wheel

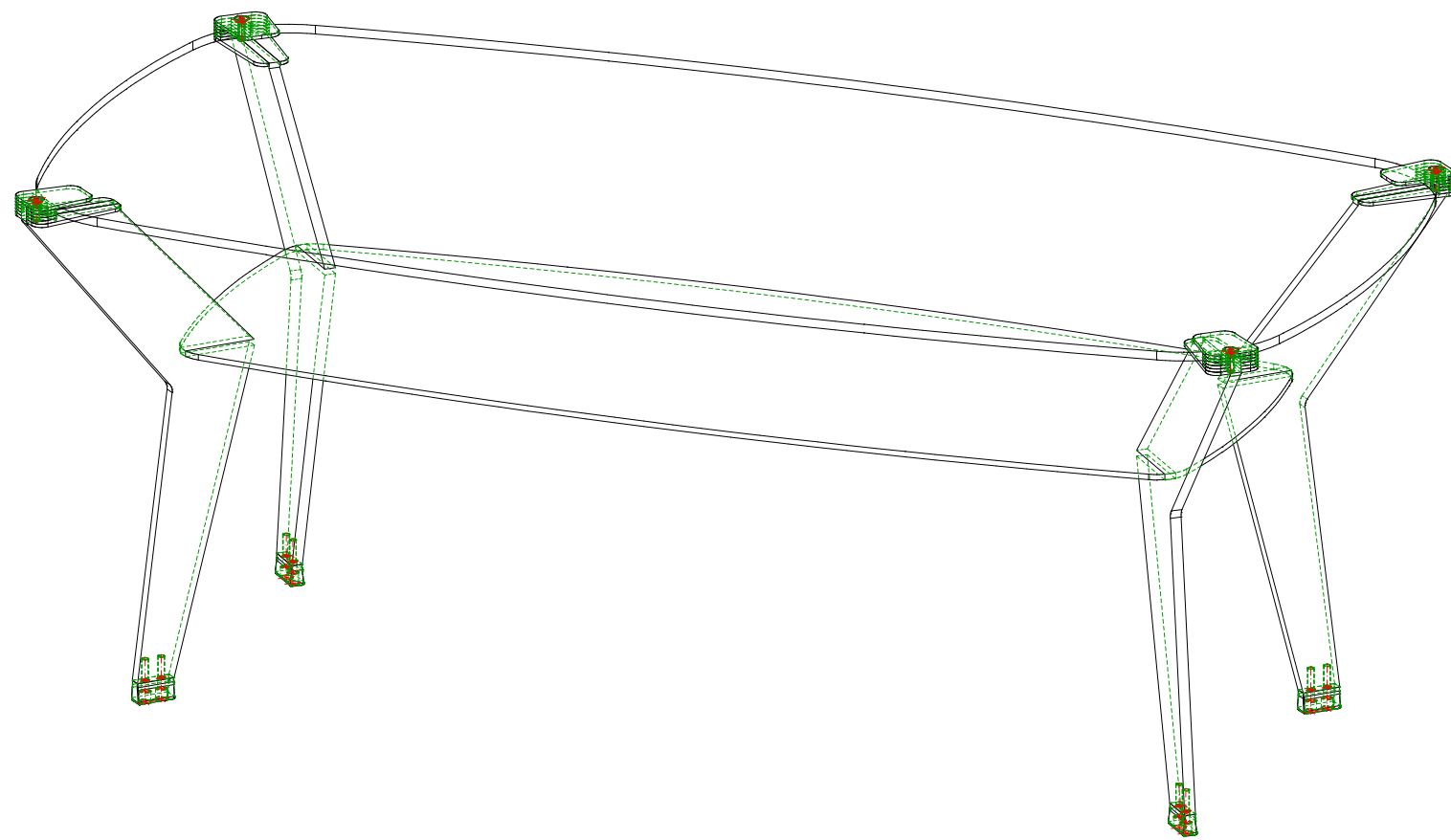
Personal Project _ Designed & Built



05.3 SELECTED WORKS

Coffee Table

Personal Project _ Designed & Built



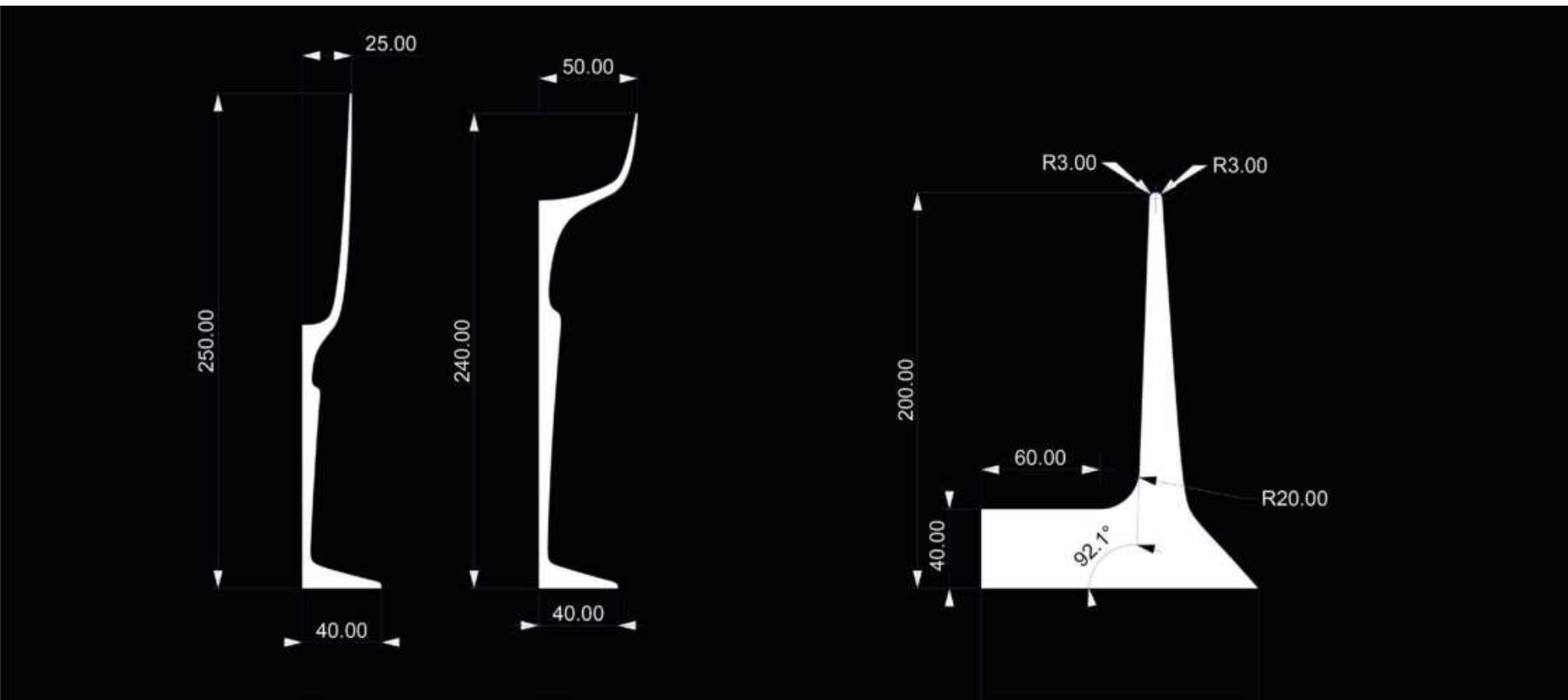
05.4 SELECTED WORKS

Spiegelau Design Language Exploration

A glassware concept developed by studying Spiegelau’s existing design language and translating its characteristic proportions, silhouettes and visual identity into a new product.

- Design Language Analysis
- Form Development
- Proportion & Silhouette
- Glassware Design
- Product Visualization

Academic Design Exercise · Unaffiliated with Spiegelau



More Projects

to be explored online

behance.net/bedirhanckr
bedirhancakiroglu.com



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Thanks

Let's have a chat.